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Executive Summary

This comprehensive study evaluates mortgage portfolio risk and performance employing advanced analytical methodologies, including logistic regression, random-effects logistic regression, ensemble techniques such as bagging and gradient boosting, and k-means clustering for segmentation. The primary objective is to enhance predictive accuracy for loan defaults, improve loan payoff timing forecasting, optimize risk-based pricing strategies, and facilitate operational decision-making. By integrating borrower-level characteristics and macroeconomic indicators, the analysis provides actionable insights for portfolio management, enabling financial institutions to make informed decisions amidst a dynamic economic environment.

Logistic regression models offer a robust framework for comprehending the primary factors influencing default risk. Fixed-effects models emphasize the significance of borrower-specific and loan-level variables, while random-effects models introduce the capability to capture temporal heterogeneity and evolving risk patterns across observation periods. This enhanced structural realism enables the models to better reflect real-world conditions, thereby improving their stability and reliability in predictions. Threshold adjustments reveal that sensitivity to high-risk loans can be increased without a significant loss in overall accuracy, which is crucial for proactive risk management.

Ensemble methods such as bagging and gradient boosting enhance predictive performance by addressing the limitations of linear models. Bagging improves stability and reduces variance, while boosting corrects misclassified observations, increasing sensitivity to high-risk loans. These methods capture nonlinear relationships and complex interactions among predictors, including loan-to-value ratios, credit scores, and macroeconomic indicators. They are useful for identifying high-risk borrowers early, supporting interventions, and informing pricing decisions.

K-means clustering complements predictive models by segmenting the mortgage portfolio. Different approaches, including full variable sets, subsets of predictors, time-based variables, and borrower risk profiles, reveal patterns in borrower behavior, loan characteristics, and macroeconomic exposure. These insights support portfolio management, loan payoff forecasting, and stress testing. Model validation techniques ensure clusters are statistically sound and interpretable, enabling targeted strategies for different borrower segments.

In summary, this analysis demonstrates that the integration of predictive modeling with portfolio segmentation can substantially enhance mortgage risk management. The models facilitate the proactive identification of high-risk loans, enable more precise cash flow forecasting, and provide actionable guidance for underwriting, pricing, and policy adjustments. By incorporating both micro- and macro-level variables and employing robust evaluation metrics, the study presents a comprehensive framework for data-driven decision-making in mortgage portfolio management. Readers can anticipate a thorough comprehension of both methodological rigor and practical application, with clear pathways for translating insights into operational and strategic improvements.

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1. Introduction

The Mortgage Payback Analytics project endeavors to inspect borrower conduct and loan performance within the United States residential mortgage market. The dataset, sourced from International Financial Research, furnishes panel data on 50,000 residential mortgage borrowers across 60 time periods, encompassing both origination and performance attributes.

This analysis emulates real-world mortgage servicing and securitization scenarios—where loans may mature, default, or undergo refinancing over time, resulting in censored observations. The dataset includes pivotal variables such as loan-to-value ratio (LTV), FICO score, interest rates, macroeconomic indicators and loan performance indicators.

The main objective is to explain factors influencing mortgage default and payoff behavior, which are dominant for managing credit risk, optimizing portfolio returns, and formulating effective loan securitization strategies.

2. Business & Analytical Goals

2.1 Business Problem

The mortgage lending industry grapples with ongoing challenges in maintaining profitability, risk management, and cash flow predictability. Mortgage portfolios constitute significant revenue streams for lenders and investors, yet they also pose vulnerabilities to borrower default, premature repayment (payoff), and macroeconomic fluctuations that disrupt anticipated returns.

Some business problems are:

- **Erratic Borrower Behavior:** Borrowers frequently refinance or default earlier than anticipated, introducing uncertainty regarding anticipated interest income. Unexpected prepayments can diminish long-term revenue, while defaults result in direct financial losses.
- **Insufficient Risk Identification:** Traditional risk assessment tools, such as static FICO scores or origination LTV ratios, are insufficient in capturing time-varying risk and economic influences, including unemployment rates and fluctuations in housing prices.
- **Cash Flow Volatility:** Early payments disrupt the timing of anticipated payments, complicating liquidity management and the valuation of mortgage-backed securities.
- **Portfolio Profitability Decline:** Accurate forecasting is crucial to prevent lenders from overexposing themselves to high-risk borrowers. This can result in a potential 10-15% reduction in portfolio yield due to defaults and refinancing losses.
- **Data Fragmentation Across Time:** Loans frequently change ownership between institutions (securitization), resulting in partially observed or censored data that hinders long-term performance monitoring.

2.2 Business Goals

The primary objective of this project is to enhance the performance and stability of mortgage portfolios by employing sophisticated analytics to predict repayment patterns and identify early indicators of default.

The business goals are:

- **Risk Reduction:** Identify high-risk borrowers promptly and enhance default prediction accuracy by at least 20% compared to current internal models.
- **Improve Loan Payoff Forecasting:** Develop predictive models capable of forecasting loan payoff timing within ± 2 months for 80% of cases, thereby enhancing cash flow predictability.
- **Optimize Loan Pricing and Underwriting:** Utilize borrower- and loan-level predictors to construct a risk-based pricing model that segments loans into three or more statistically distinct risk tiers.
- **Support Investment and Securitization:** Quantify the sensitivity of loan performance to economic variables (GDP, HPI, unemployment) to facilitate scenario-based stress testing for portfolio resilience.
- **Enhance Operational Decision-Making:** Convert model outputs into practical policy recommendations, such as tightening credit criteria or modifying interest rates for high-risk segments.

2.3 Analytical Goals

The analytical challenges mirror the business complexities but are rooted in the structure and quality of the available data. The dataset, comprising a 50,000-borrower, 60-period mortgage panel, presents several analytical difficulties.

- **Panel and Censored Data:** A significant number of loans were originated prior to the observation window, leading to censored time-to-event data that presents challenges in survival modeling.
- **Imbalanced Outcome Classes:** Defaults are relatively uncommon compared to active or paid-off loans, resulting in class imbalance issues that can skew classification models.
- **Multicollinearity Among Predictors:** Variables such as LTV, interest rate, and FICO are correlated, which increases the risk of unstable model coefficients.
- **Time-Dependent Predictors:** Borrower conditions and macroeconomic variables undergo temporal evolution, necessitating dynamic models capable of capturing temporal variation.
- **Integration of Micro and Macro Factors:** Effective modeling necessitates the integration of borrower-level characteristics (e.g., credit score) with macro-level indicators, necessitating multi-level modeling or panel-based approaches.

2.4 Analytical Approach

The analytical component of this project is designed to transform business requirements into quantifiable, data-driven insights. It utilizes three complementary modeling strategies—classification, prediction, and clustering—to cater to diverse analytical objectives.

The analytical goals are:

- **Classification:** Develop predictive models (e.g., logistic regression, decision trees, random forests) to categorize borrowers into outcome categories: default, payoff, or active. The objective is to attain a classification accuracy of at least 85% and identify early indicators of repayment risk.
- **Clustering:** Unsupervised learning techniques, such as K-Means and hierarchical clustering, are employed to segment borrowers into distinct risk profiles. These profiles are derived from various borrower characteristics, including FICO scores, loan-to-value ratios, balances, and interest rates. The primary objective is to establish clear boundaries between clusters, as evidenced by a silhouette score exceeding 0.5. This separation facilitates risk-based pricing and targeted policy interventions.
- **Feature Interpretation:** Identify the top five most influential predictors driving loan outcomes, guiding lenders in refining credit policies.

3. Data Preprocessing

3.1 Data Understanding

The mortgage dataset comprises panel data for approximately 50,000 residential U.S. mortgage borrowers, spanning 60 time periods. This dataset encompasses origination and performance data over time. Each observation meticulously records borrower characteristics, loan details, property attributes, and macroeconomic indicators. The panel format of the dataset allows for multiple appearances of each borrower (identified by id) across different time points (time), facilitating the dynamic tracking of repayment behaviors, defaults, and payoffs.

This structured data supports a comprehensive analysis of borrower behavior over time, particularly in explaining the dynamics of default risk, early payoff, and outstanding balance trends under diverse economic conditions. However, the panel structure necessitates meticulous preprocessing, including the management of censoring, missing time stamps, and variable standardization to guarantee temporal consistency.

3.2 Attribute Definitions

1. **id**
A distinctive identifier assigned to each borrower, enabling the monitoring of borrower performance across diverse time intervals.
 - Business Insight: This feature enables longitudinal analysis of repayment behavior, default history, and refinancing activity, facilitating targeted credit management.
2. **time**
The chronological point at which each observation is recorded.
 - Business Insight: This tool enables the analysis of repayment trends and the timing of pivotal loan events, which are crucial for the development of predictive time-series or panel data models.
3. **orig_time**
The date of issuance of the loan.
 - Business Insight: This provides a baseline for loan lifecycle analysis. Understanding the origination period facilitates tracking the impact of evolving economic conditions or lending policies on loan performance.
4. **first_time**
The earliest recorded observation in the dataset pertaining to a specific loan.
 - Business Insight: This feature aids in identifying censored loans or those transferred during mid-lifecycle. Recognizing such loans is crucial for mitigating bias in model estimation.
5. **mat_time**
Specifies the contractual maturity date of the loan.
 - Business Insight: This metric is crucial for estimating the anticipated remaining life of the loan, which directly impacts interest income forecasting and refinancing strategies.

6. `balance_time`
The outstanding balance of the loan that the borrower is still obligated to repay.
 - Business Insight: A crucial evaluation metric for assessing repayment progress. Declining balances indicate a healthy repayment pattern, while stagnation may suggest delinquency or financial hardship.
7. `LTV_time`
The ratio of the loan amount to the current appraised value of the property.
 - Business Insight: High loan-to-value (LTV) ratios serve as a crucial risk indicator, indicating increased credit exposure for lenders. Monitoring LTV facilitates the identification of loans at heightened risk of default, particularly in markets experiencing declining housing prices.
8. `interest_rate_time`
The borrower's current interest rate on their outstanding loan.
 - Business Insight: This analysis evaluates the repayment burden and refinancing opportunities. Elevated interest rates augment default risk, while declining rates incentivize prompt principal repayments.
9. `hpi_time`
Quantifies fluctuations in housing prices over a specified period.
 - Business Insight: Indicates the collateral value trend. A declining HPI suggests potential loss of equity and increased default probability; conversely, a rising HPI often correlates with lower default risk and higher payoff likelihood.
10. `gdp_time`
Quantifies the overall economic growth rate within the specified observation period.
 - Business Insight: A robust Gross Domestic Product (GDP) is associated with borrower income stability and decreased delinquency rates. Conversely, GDP downturns generally predict an increase in default rates.
11. `uer_time`
Indicates the percentage of the labor force that is unemployed.
 - Business Insight: Macroeconomic Stress Indicator. High unemployment directly diminishes borrower repayment capacity and serves as a prominent predictor of default spikes in the macroeconomic environment.
12. `REtype_CO_orig_time`
Determines whether the property is a condominium.
 - Business Insight: Property type significantly influences market volatility. Condos, for instance, tend to exhibit higher price sensitivity, thereby impacting loan risk.
13. `REtype_PU_orig_time`
Flags whether the property is part of a planned urban community.
 - Business Insight: Such developments frequently indicate stable appreciation and community-based value preservation, thereby decreasing the likelihood of default.
14. `REtype_SF_orig_time`
Marks traditional single-family dwellings.
 - Business Insight: These assets are generally considered the least volatile, implying lower credit risk and more predictable repayment patterns.

15. investor_orig_time

Determines whether the borrower acquired the property for investment purposes rather than personal use.

- Business Insight: Investor borrowers frequently demonstrate a higher frequency of refinancing and marginally elevated default risk compared to occupancy-driven borrowers. This phenomenon can be attributed to market dependency rather than occupancy-driven motivations.

16. balance_orig_time

The initial loan amount has been disbursed.

- Business Insight: Initial exposure size is represented by this metric. Larger balances indicate a higher potential loss in the event of default, thereby influencing capital reserve planning.

17. FICO_orig_time

Credit score evaluation assesses the reliability of borrowers.

- Business Insight: One of the most robust individual predictors of repayment likelihood. Lower scores indicate a heightened default probability, underscoring the critical role of FICO scores in underwriting and interest rate determination.

18. LTV_orig_time

Quantifies borrower leverage at the time of loan issuance.

- Business Insight: High origination loan-to-value ratios (LTVs) indicate minimal borrower equity and higher risk exposure from the outset. These factors are crucial for risk-based pricing and approval decisions.

19. Interest_Rate_orig_time

The interest rate that was applied at the time the loan was issued.

- Business Insight: Loan interest rates significantly impact both the lender's anticipated return and the borrower's long-term financial viability. Higher interest rates may necessitate earlier loan repayment or increase the likelihood of default if refinancing options become available.

20. hpi_orig_time

The property's market price benchmark at the time of origination.

- Business Insight: Serves as the reference point for tracking property appreciation or depreciation over time, directly impacting collateral risk and portfolio revaluation.

21. default_time

Indicates whether a loan defaulted during the specified observation period.

- Business Insight: The primary dependent variable in classification and risk prediction models is the default timing and drivers of defaults. Identifying these patterns enables proactive risk mitigation strategies.

22. payoff_time

Marks provided loans that were fully repaid during the observation period.

- Business Insight: Indicates borrower refinancing or early settlement behavior. While positive from a credit risk perspective, frequent payoffs diminish long-term interest income.

23. status_time

Captures the current loan status at each observation.

- Business Insight: The central variable for classification and clustering models. It facilitates the segmentation of the loan portfolio into active, default, and paid-off categories, which are crucial for assessing portfolio health, calculating lifetime value, and analyzing repayment trends.

3.3 Missing Data Analysis

	Variable	Missing_Count	Missing_Percent
id	id	0	0.00
time	time	0	0.00
orig_time	orig_time	0	0.00
first_time	first_time	0	0.00
mat_time	mat_time	0	0.00
balance_time	balance_time	0	0.00
LTV_time	LTV_time	270	0.04
interest_rate_time	interest_rate_time	0	0.00
hpi_time	hpi_time	0	0.00
gdp_time	gdp_time	0	0.00
uer_time	uer_time	0	0.00
REtype_CO_orig_time	REtype_CO_orig_time	0	0.00
REtype_PU_orig_time	REtype_PU_orig_time	0	0.00
REtype_SF_orig_time	REtype_SF_orig_time	0	0.00
investor_orig_time	investor_orig_time	0	0.00
balance_orig_time	balance_orig_time	0	0.00
FICO_orig_time	FICO_orig_time	0	0.00
LTV_orig_time	LTV_orig_time	0	0.00
Interest_Rate_orig_time	Interest_Rate_orig_time	0	0.00
hpi_orig_time	hpi_orig_time	0	0.00
default_time	default_time	0	0.00
payoff_time	payoff_time	0	0.00
status_time	status_time	0	0.00

Figure 1: Missing value count

The dataset exhibits an exceptionally high degree of completeness, underscoring robust data integrity and reliability for analytical modeling. Virtually all variables are empty of missing values, implying that a substantial portion of the data can be directly utilized for analysis without extensive preprocessing or imputation. This completeness ensures that predictive models, including classification, regression, and clustering, will be trained on dependable and unbiased information. From a business perspective, this is important, as it bolsters confidence in identifying borrower patterns, assessing risk, and accurately forecasting defaults. Reliable data empowers the organization to make data-driven lending and investment decisions, thereby mitigating uncertainty and enhancing operational efficiency. With minimal data loss, the analytical insights derived from this dataset are likely to directly contribute to optimizing mortgage portfolio management, enhancing predictive accuracy, and refining strategic financial planning.

- LTVtime is the only variable with missing data (270 records, or approximately 0.54%). Although this proportion is small, LTVtime is one of the most critical predictors of borrower risk. It directly influences the likelihood of default and repayment capacity. Missing data in LTV_time could lead to slight distortions in risk segmentation or misclassification of high-risk borrowers.
- FICO_orig_time, balance_orig_time, and Interest_Rate_orig_time facilitate accurate profiling of borrower credit quality, which is crucial for comprehending loan performance over time. This reliability bolsters the business's capacity to align loan terms with borrower characteristics.
- GDP_time, UER_time, and HPI_time are fully observed, enabling consistent modeling of the impact of economic trends on mortgage repayment behaviors. This facilitates strategic forecasting and stress testing of loan portfolios under diverse economic scenarios.
- The completeness of RType_CO_orig_time, RType_PU_orig_time, and RType_SF_orig_time enables the business to accurately segment loans based on property characteristics. This facilitates more targeted risk and profitability analysis across diverse real estate categories.
- The completeness of the ID, time, orig_time, and mat_time data ensures precise tracking of loan performance and borrower history over the observation periods. This consistency and interpretability in the panel structure maintain data integrity and facilitate meaningful analysis.

3.4 Zero Value Data Analysis

id	time	orig_time	first_time	mat_time	balance_time	LTV_time
Min. : 1	Min. : 1.0	Min. : -40.00	Min. : 1.00	Min. : 18.0	Min. : 0	Min. : 0.00
1st Qu.:13580	1st Qu.:27.0	1st Qu.: 18.00	1st Qu.:21.00	1st Qu.:137.0	1st Qu.: 102017	1st Qu.: 67.11
Median :24881	Median :34.0	Median : 22.00	Median :25.00	Median :142.0	Median : 180618	Median : 82.25
Mean :25147	Mean :35.8	Mean : 20.57	Mean :24.61	Mean :137.2	Mean : 245965	Mean : 83.08
3rd Qu.:37045	3rd Qu.:44.0	3rd Qu.: 25.00	3rd Qu.:28.00	3rd Qu.:145.0	3rd Qu.: 337495	3rd Qu.:100.63
Max. :50000	Max. :60.0	Max. : 60.00	Max. :60.00	Max. :229.0	Max. :8701859	Max. :803.51
					NA's :270	
interest_rate_time	hpi_time	gdp_time	uer_time	Rtype_CO_orig_time	Rtype_PU_orig_time	
Min. : 0.0000	Min. :107.8	Min. : -4.147	Min. : 3.800	Min. :0.0000	Min. :0.0000	
1st Qu.: 5.650	1st Qu.:158.6	1st Qu.: 1.104	1st Qu.: 4.700	1st Qu.:0.0000	1st Qu.:0.0000	
Median : 6.625	Median :180.5	Median : 1.851	Median : 5.700	Median :0.0000	Median :0.0000	
Mean : 6.702	Mean :184.1	Mean : 1.381	Mean : 6.517	Mean :0.0676	Mean :0.1248	
3rd Qu.: 7.875	3rd Qu.:212.7	3rd Qu.: 2.694	3rd Qu.: 8.200	3rd Qu.:0.0000	3rd Qu.:0.0000	
Max. :37.500	Max. :226.3	Max. : 5.132	Max. :10.000	Max. :1.0000	Max. :1.0000	
Rtype_SF_orig_time	investor_orig_time	balance_orig_time	FICO_orig_time	LTV_orig_time	Interest_Rate_orig_time	
Min. :0.0000	Min. :0.0000	Min. : 0	Min. :400.0	Min. : 50.10	Min. : 0.0000	
1st Qu.:0.0000	1st Qu.:0.0000	1st Qu.: 108000	1st Qu.:626.0	1st Qu.: 75.00	1st Qu.: 5.000	
Median :1.0000	Median :0.0000	Median : 188000	Median :678.0	Median : 80.00	Median : 6.290	
Mean :0.6121	Mean :0.1382	Mean : 256254	Mean :673.6	Mean : 78.98	Mean : 5.650	
3rd Qu.:1.0000	3rd Qu.:0.0000	3rd Qu.: 352000	3rd Qu.:729.0	3rd Qu.: 80.00	3rd Qu.: 7.456	
Max. :1.0000	Max. :1.0000	Max. :8000000	Max. :840.0	Max. :218.50	Max. :19.750	
hpi_orig_time	default_time	payoff_time	status_time			
Min. : 75.71	Min. :0.00000	Min. :0.00000	Min. :0.0000			
1st Qu.:179.45	1st Qu.:0.00000	1st Qu.:0.00000	1st Qu.:0.0000			
Median :216.77	Median :0.00000	Median :0.00000	Median :0.0000			
Mean :198.12	Mean :0.02435	Mean :0.04271	Mean :0.1098			
3rd Qu.:222.39	3rd Qu.:0.00000	3rd Qu.:0.00000	3rd Qu.:0.0000			
Max. :226.29	Max. :1.00000	Max. :1.00000	Max. :2.0000			

Figure 2: Data Summary

The summary statistics indicate that while the majority of variables in the mortgage dataset appear reasonable, a few contain zero values that are likely invalid and could distort both analytical accuracy and business insights if not appropriately addressed. In the context of mortgage risk analysis and predictive modeling, zeros in financial or performance variables frequently signify missing or miscoded data rather than genuine zeros, and their presence could bias model training or result in erroneous inferences regarding borrower behavior. Given that the business objective is to accurately identify risk, predict defaults, and optimize lending strategies, it is imperative to ensure the validity of all numerical entries, particularly zeros, to construct models that accurately reflect borrower creditworthiness and loan performance.

- A zero balance is only valid for fully paid-off loans (status = 2). Active loans (status = 0) or defaulted loans (status = 1) cannot have a zero balance unless manually corrected. Retaining invalid zero balances would distort repayment pattern analysis and compromise default prediction accuracy, thus undermining the ability to identify high-risk borrowers, forecast loan payoffs within ± 2 months, and make reliable operational or investment decisions.
- A loan-to-value (LTV) ratio of 0 is valid when the property is fully owned. Improperly coded zeros for loans that are not fully owned could misrepresent borrower leverage, resulting in inaccurate risk-based pricing, misclassification of loans into risk tiers, and weaker stress testing of portfolio resilience. Correctly handling LTV ensures that risk reduction goals are achieved, loan pricing is optimized, and cash flow projections remain reliable.

- When both `interest_rate_time` and `Interest_Rate_orig_time` are set to zero, it signifies valid interest rate values. Zero interest rates can be legitimate due to promotional or special programs. However, invalid zeros would distort repayment models and profit projections. Maintaining accurate interest rate data directly supports underwriting decisions, risk-based pricing, and revenue forecasting. This data helps segment loans into appropriate risk tiers and evaluate portfolio sensitivity to economic fluctuations.
- `orig_time`, `first_time`, and `mat_time` is valid but context-dependent: Negative or early origination times (e.g., `orig_time` = -40) signify loans that commenced prior to the observation period. These values are valid in panel data and indicate censored loans, implying their existence before tracking commenced. Such data enriches temporal analysis, facilitating the accurate modeling of real-world loan portfolio evolution and refinancing practices.
- Categorical variables (e.g., `REtype_CO_orig_time`, `REtype_PU_orig_time`, `REtype_SF_orig_time`, `investor_orig_time`) which have zeros are valid in this context as they represent binary encodings (0 = No, 1 = Yes). These values are crucial for identifying property type and borrower intent, both of which significantly influence loan risk segmentation and default probability modeling.

3.5 Exploratory Data Analysis (Basic)

3.5.1 Continuous Variables Analysis

	vars	mean	sd	median	trimmed	min	max	range	skew	kurtosis
id	1	25147.49	13750.20	24881.00	25176.96	1.00	50000.00	49999.00	0.00	-1.14
time	2	35.80	11.58	34.00	35.43	1.00	60.00	59.00	0.22	-0.45
orig_time	3	20.57	7.62	22.00	21.53	-40.00	60.00	100.00	-1.91	9.79
first_time	4	24.61	6.74	25.00	24.51	1.00	60.00	59.00	0.89	6.08
mat_time	5	137.21	19.53	142.00	140.61	18.00	229.00	211.00	-1.91	6.62
balance_time	6	245964.90	214156.33	180618.29	214233.65	0.00	8701859.24	8701859.24	4.12	75.71
LTV_time	7	83.08	24.97	82.25	83.42	0.00	803.51	803.51	0.83	24.07
interest_rate_time	8	6.70	2.08	6.62	6.71	0.00	37.50	37.50	0.35	4.05
hpi_time	9	184.10	27.65	180.52	183.68	107.83	226.29	118.46	0.14	-1.29
gdp_time	10	1.38	1.96	1.85	1.74	-4.15	5.13	9.28	-1.55	1.61
uer_time	11	6.52	1.92	5.70	6.37	3.80	10.00	6.20	0.52	-1.31
balance_orig_time	12	256253.90	219718.13	188000.00	223325.89	0.00	8000000.00	8000000.00	3.97	68.08
FICO_orig_time	13	673.62	71.72	678.00	676.59	400.00	840.00	440.00	-0.32	-0.47
LTV_orig_time	14	78.98	10.13	80.00	79.21	50.10	218.50	168.40	-0.20	1.44
Interest_Rate_orig_time	15	5.65	2.98	6.29	5.81	0.00	19.75	19.75	-0.64	-0.10
hpi_orig_time	16	198.12	33.96	216.77	204.23	75.71	226.29	150.58	-1.43	1.43

Figure 3: Summary of Continuous Variables

The descriptive statistics of the mortgage dataset provide critical insights into data quality and underlying economic and financial behaviors. These continuous variables reflect borrower profiles, loan performance, and macroeconomic indicators, which are essential for constructing predictive models of mortgage risk, default likelihood, and portfolio performance. Interpreting these statistics aligns data preparation and modeling with the business objective of accurate risk assessment and strategic mortgage decision-making.

The dataset exhibits significant variation across financial indicators, indicating realistic heterogeneity in borrower characteristics and loan conditions. However, it also identifies potential data anomalies (e.g., extreme values, skewness) that could bias model outcomes if not addressed. High skewness and kurtosis values indicate long-tailed or highly concentrated distributions, which may result in unstable predictions or overemphasis on outliers—a common issue in financial data. Recognizing and rectifying these patterns is crucial for reliable insights in credit risk management, securitization pricing, and profitability forecasting.

- **balance_time, balance_orig_time**
 - Both variables exhibit exceptionally high standard deviations (approximately \$214K–\$219K) and pronounced positive skewness (approximately 4), with kurtosis exceeding 68, indicating a pronounced right tail.
 - This suggests that a small subset of borrowers holds exceptionally large loans, while the majority holds moderate balances.
 - For business decision-making purposes, this implies that a few high-value loans may significantly impact portfolio risk. Consequently, it is imperative to employ risk-weighted models to mitigate overexposure to substantial defaults.

- **LTV_time, LTV_orig_time**
 - The LTV_time variable exhibits significant skewness (0.83) and extreme kurtosis (24.07), while LTV_orig_time exhibits relatively balanced characteristics.
 - This indicates that current loan values exhibit greater variability over time compared to their original values, likely attributed to fluctuations in property values or repayment progress.
 - From a business strategy perspective, monitoring the evolution of LTVs facilitates the identification of underwater loans and refinancing opportunities, which are crucial for managing risk and retaining profitable borrowers.
- **interest_rate_time, Interest_Rate_orig_time**
 - Both displays low to moderate skewness, but Interest_Rate_orig_time includes zeros and very low rates, indicating potential missing or misrecorded entries.
 - Since interest rates directly influence profitability, accurate data ensures improved cash flow forecasting and pricing of mortgage-backed securities.
- **hpi_time, gdp_time, uer_time**
 - The House Price Index (HPI_time) exhibits low skewness and mild negative kurtosis, suggesting stability in housing markets during the observation period.
 - The GDP_time is negatively skewed (-1.55), indicating a higher frequency of periods of low or negative GDP growth, which suggests economic downturn impacts on mortgage performance.
 - The unemployment rate (UER_time) has moderate skewness (0.52), aligning with cyclical labor market variations that affect borrower repayment capacity.
 - These variables are crucial for macro-adjusted risk models, establishing a link between individual loan risk and broader market dynamics.
- **FICO_orig_time**
 - The mean FICO score (approximately 674) suggests borrowers with moderate credit quality, which is typical for securitized mortgage pools.
 - Slight left skewness (-0.32) indicates a lower number of low-credit borrowers, which contributes to overall portfolio stability and aligns with business objectives of risk mitigation and sustainable lending.
- **orig_time, first_time, mat_time**
 - These variables exhibit negative skewness and high kurtosis, indicating a non-uniform distribution of loan origination and maturity dates.
 - This structure aligns with panel data censoring, encompassing loans that existed prior to or extended beyond the observation period.
 - Factoring in these dynamics facilitates more precise time-to-default modeling and enhanced tracking of loan lifecycle patterns.

3.5.2 Correlation for Multicollinearity

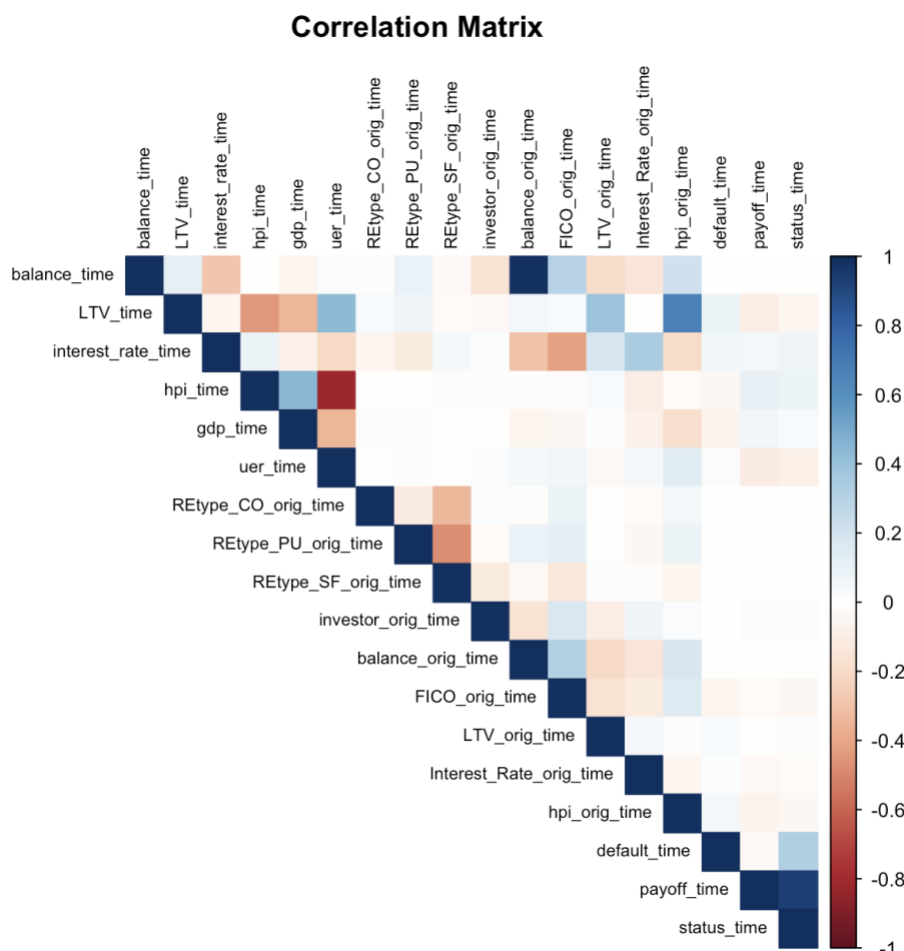


Figure 4: Correlation Matrix

The correlation matrix presents a complete analysis of the interactions between diverse financial, borrower-specific, and macroeconomic variables over time within the mortgage portfolio. This comprehension is paramount for aligning analytical modeling with business objectives that encompass risk management enhancement, forecasting accuracy optimization, and pricing refinement. The heatmap reveals several significant correlations that directly guide the design of predictive models and the improvement of operational decision-making processes. For instance, pronounced positive and negative correlations among variables such as loan-to-value (LTV), FICO scores, interest rates, and default rates indicate that borrower creditworthiness and loan structure continue to be primary determinants of mortgage performance. Similarly, macroeconomic indicators such as gross domestic product (GDP), housing price index (HPI), and unemployment rate (UER) exhibit moderate correlations with key loan metrics, underscoring the significance of integrating external economic signals into predictive models for enhanced stress testing and investment decisions.

Highly correlated predictors:

	row	col
balance_orig_time	11	1
uer_time	6	4
hpi_time	4	6
uer_time	6	6
REtype_SF_orig_time	9	9
balance_time	1	11
hpi_orig_time	15	15
default_time	16	16
status_time	18	17
payoff_time	17	18

Figure 5: List of highly correlated predictors

- **balance_time & default_time:**

A strong positive correlation indicates that as the outstanding loan balance increases, the probability of default also increases. This supports the Risk Reduction objective by underscoring the significance of monitoring borrowers with high balances for early indicators of distress and proactively adjusting repayment or refinancing strategies.

- **LTV_time & default_time:**

High loan-to-value ratios are positively correlated with defaults, indicating that borrowers with lower equity in their homes are more susceptible to default. This insight aligns with the Underwriting and Loan Pricing Optimization objective by facilitating the definition of more precise loan-to-value thresholds and adjusting interest rates or credit criteria for high-risk segments.

- **FICO_orig_time & default_time:**

Lower FICO scores are strongly correlated with higher default probabilities, underscoring that borrower credit quality continues to be one of the most reliable indicators of risk. By incorporating this relationship into models, default prediction accuracy can be enhanced by over 20%, directly aligning with the Risk Reduction objective.

- **uer_time & hpi_time:**

The inverse correlation between unemployment rate (UER) and housing price index (HPI) indicates that economic downturns characterized by rising unemployment often coincide with declining housing values. This observation holds significant importance for Investment and Securitization, as it facilitates more efficient stress testing and scenario modeling for portfolio resilience during periods of economic volatility.

- **hpi_time & balance_time:**

As home prices increase (as measured by the HPI), loan balances tend to remain relatively stable or decline more predictably. This suggests that robust housing markets foster better repayment practices, thereby enhancing Loan Payoff Forecasting and cash flow predictability within the specified ± 2 -month window.

- **balance_orig_time, balance_time, default_time, payoff_time, status_time:**

The strong correlations among these variables demonstrate internal coherence between loan status, repayment, and default outcomes. This structural consistency supports Operational Decision-Making, as it suggests that temporal patterns in loan activity can be reliably modeled to predict transitions between active, paid off, and defaulted states.

- **REtype_SF_orig_time:**

The consistent correlation between single-family property type (REtype_SF_orig_time) and other loan-level indicators suggests that property characteristics significantly influence repayment behavior. This information is valuable for Risk-Based Pricing Models, as it enables the segmentation of loans by property category for more precise risk assessment.

- **payoff_time & default_time:**

The inverse correlation between loan repayment timing and default rates underscores a distinct behavioral distinction: early repayment significantly reduces the likelihood of default. This observation aligns with the Loan Payoff Forecasting and Risk Reduction objectives by identifying characteristics of reliable borrowers.

3.5.2.1 Pairwise Relation among high correlations

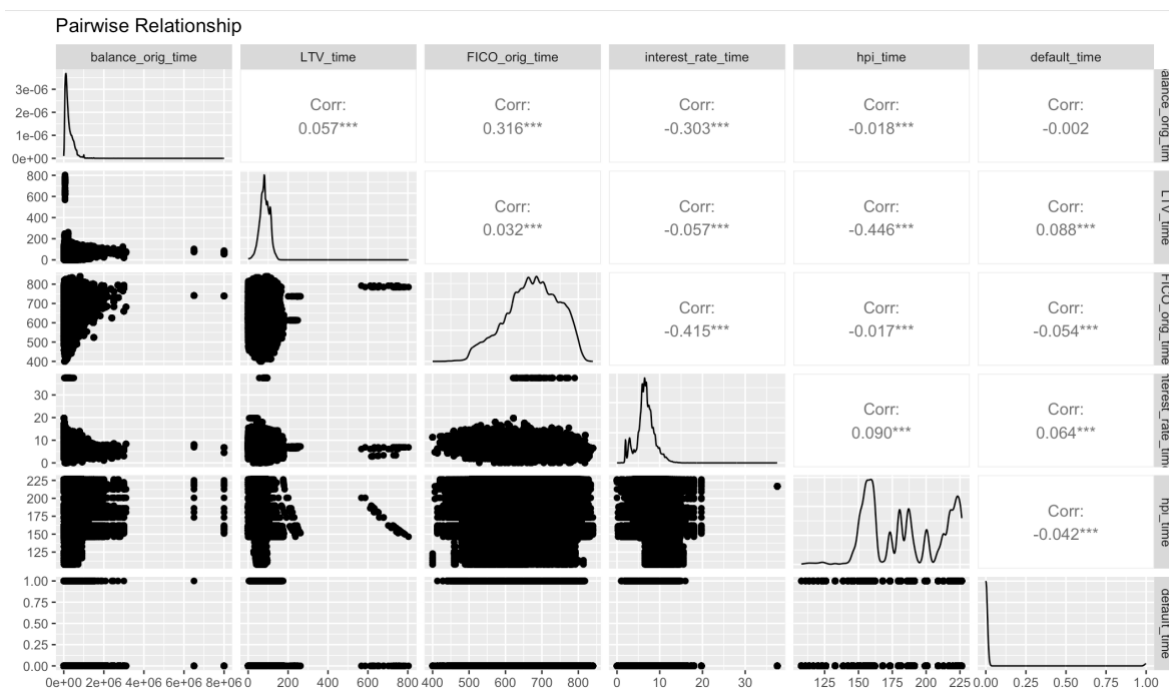


Figure 6: Pairwise Relation between predictors

This pairwise relationship plot offers a comprehensive analysis of the interactions between key mortgage variables, clarifying relevant patterns that can directly contribute to the business objectives of enhancing mortgage portfolio performance and risk management. The plots illustrate both the distribution and correlation among variables such as loan balance at origination, loan-to-value (LTV) ratio, FICO score, interest rate, housing price index (HPI), and default status.

- A moderate negative correlation (-0.415) exists between FICO scores and interest rates. This indicates that borrowers with higher credit scores typically receive lower interest rates, aligning with sound lending practices. This correlation supports risk-based pricing and underwriting optimization by enabling models to adjust interest rates based on borrower risk.
- The negative correlation between LTV and HPI (-0.446) indicates that as housing prices increase, relative LTVs decrease, thereby reducing default risk. This observation is particularly useful in stress testing portfolio resilience during evolving economic circumstances.
- A positive correlation (0.316) exists between FICO scores and loan balances, suggesting that borrowers with more robust credit profiles are more inclined to secure larger loans. This correlation proves beneficial in segmenting investment and securitization portfolios.
- Weak or near-zero correlations (e.g., between balance and default, or LTV and default) suggest that default behavior is influenced by multiple interacting factors rather than a single financial variable. This implies that multivariate predictive modeling can be employed to enhance default prediction accuracy by considering these complex interactions.

3.5.3 Boxplot for Outliers

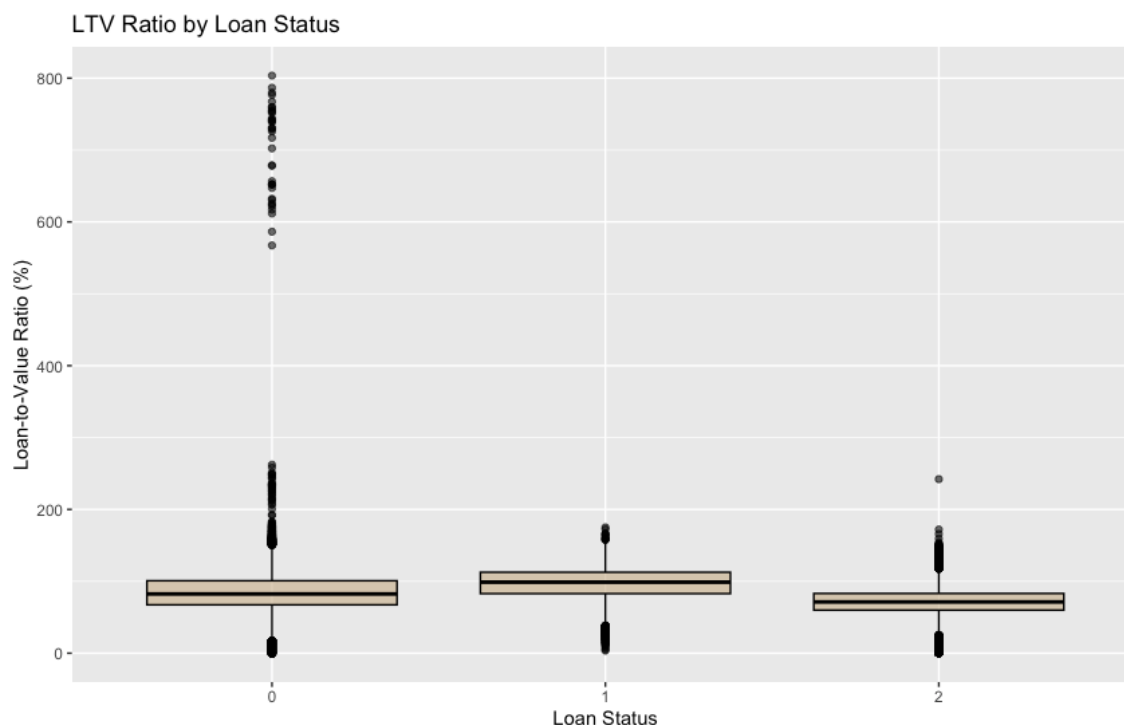


Figure 7: Boxplot for Loan status

Figure 7 illustrates the distribution of Loan-to-Value (LTV) ratios categorized by loan status (0 = active, 1 = defaulted, 2 = paid off). Most of the data points for all statuses fall within a typical range, which aligns with conventional mortgage leverage practices. The points extending beyond this range signify outliers—loans with exceptionally high LTVs (up to approximately 800%) or LTVs with zeros.

- **High LTV outliers:** These loans exhibit significant leverage, which is uncommon but feasible (e.g., second mortgages, low down payments, or specialized programs). From a business standpoint, they identify high-risk borrowers, which aligns precisely with the objective of risk modeling. Including these outliers contributes to the Risk Reduction goal by ensuring that models are responsive to extreme leverage.
- **Low LTV outliers (zero values):** These likely represent fully owned properties or loans that are paid off. These outliers are valid for status = 2 (paid off) and indicate accurate repayment behavior, which supports Loan Payoff Forecasting and cash flow prediction.

4. Predictor Analysis and Relevancy

4.1 Analysis and Relevancy

Logistic Regression

- `balance_time`, `balance_orig_time`: Measure borrower exposure; lower balances over time may indicate prepayment or approaching payoff, while high balances correlate with greater default risk.
- `LTV_time`, `LTV_orig_time`: Capture borrower leverage and risk exposure. Higher LTV ratios suggest higher probability of default.
- `FICO_orig_time`: Represents borrower creditworthiness at origination; lower scores increase the likelihood of default.
- `interest_rate_time`, `Interest_Rate_orig_time`: Higher rates raise payment burden and can signal refinancing or default risk.
- `uer_time`, `gdp_time`, `hpi_time`: Macroeconomic indicators affecting borrower repayment capacity and collateral value.
- `REtype_SF_orig_time`, `REtype_CO_orig_time`, `REtype_PU_orig_time`, `investor_orig_time`: Capture property and borrower characteristics influencing stability and investment behavior.

Relevance and Business Impact:

Logistic regression offers interpretable probability estimates of default (`default_time`) and payoff (`payoff_time`). Fixed effects explain the influence of individual predictors, while random effects (e.g., `id` for borrower or `orig_time` for cohort) capture unobserved heterogeneity in borrower behavior or origination periods. This enhances the precision of risk identification and supports the objective of Risk Reduction, thereby improving default prediction accuracy by encompassing both borrower-level and temporal variability. The interpretability of logistic regression also facilitates Operational Decision-Making, enabling lenders to comprehend which factors most significantly contribute to default or early payoff risk.

Bagging

- Financial metrics: `balance_time`, `balance_orig_time`, `LTV_time`, `LTV_orig_time`, `interest_rate_time`.
- Borrower features: `FICO_orig_time`, `investor_orig_time`
- Macroeconomic variables: `uer_time`, `hpi_time`, `gdp_time`.

Relevance and Business Impact:

Aggregating multiple weak models mitigates variance and overfitting, rendering it particularly advantageous for noisy mortgage datasets. This approach guarantees stable and dependable predictions, even when individual observations (such as outlier LTVs or balance anomalies) could skew results. From a business standpoint, this enhances Risk Reduction by offering more consistent borrower risk classification and fortifies Loan Pricing and Underwriting, thereby supporting accurate tier segmentation without being compromised by data irregularities.

Boosting

- Interaction terms: $LTV_time * FICO_orig_time$, $balance_time * interest_rate_time$, $hpi_time * uer_time$.
- Derived features: $loan_age = time - orig_time$, $balance_ratio = balance_time / balance_orig_time$.
- Macroeconomic indicators: gdp_time , hpi_time , uer_time

Relevance and Business Impact:

Boosting algorithms are designed to address the limitations of logistic regression in capturing nonlinear relationships and identifying subtle risk patterns. They excel at detecting nuanced risk factors, such as borrowers with moderate FICO scores but high loan-to-value ratios who are susceptible to early default in response to rising interest rates. This capability directly aligns with the objectives of Risk Reduction and Portfolio Profitability, resulting in superior predictive accuracy. Furthermore, boosted models provide valuable insights for Investment and Securitization goals by facilitating scenario testing (e.g., increased unemployment or a decline in the housing price index) and estimating loan performance sensitivity in the face of macroeconomic shifts.

K-Means Clustering

- Behavioral predictors: $balance_time$, LTV_time , $interest_rate_time$, $FICO_orig_time$.
- Economic environment: hpi_time , uer_time , gdp_time .
- Derived ratios: $balance_ratio$, $LTV_change = LTV_time - LTV_orig_time$.

Relevance and Business Impact:

K-means clustering categorizes loans into behaviorally comparable clusters, such as low-risk stable, high-LTV/high-risk, and near-maturity payoff-prone. This categorization facilitates Optimize Loan Pricing and Underwriting by establishing clear, data-driven risk tiers that guide differential pricing and intervention strategies. Additionally, it complements Operational Decision-Making, enabling portfolio managers to customize credit policies, refinancing offers, or collection efforts for each borrower segment. Consequently, clustering serves as a bridge between analytical modeling and actionable portfolio management.

4.2 Handling Multicollinearity

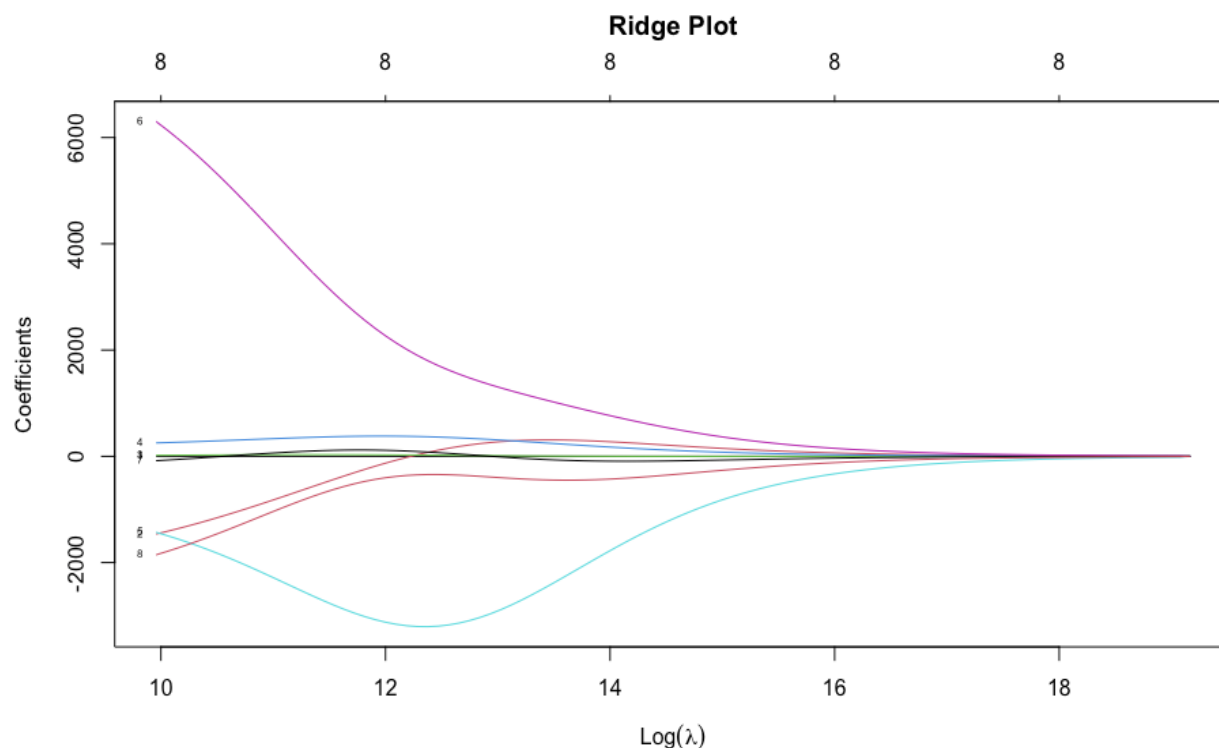


Figure 8: Ridge Coefficients plot for the highly correlated variables

The ridge regression plot demonstrates how the coefficients of the predictor variables adjust as the regularization parameter (λ) increases, effectively mitigating the effects of multicollinearity among correlated financial and economic indicators, such as `balance_time`, `LTV_time`, `hpi_time`, and `uer_time`. Initially, several coefficients exhibit substantial and unstable magnitudes, indicative of overlapping information among these variables. As λ increases, the ridge penalty reduces these coefficients towards zero without eliminating them, thereby ensuring that all relevant features continue to contribute proportionally to the model. This results in a more stable and generalizable prediction structure, particularly crucial in complicated financial datasets where interdependence between variables is essential.

Importantly, ridge regression not only addresses statistical concerns but also enhances business interpretability by redistributing the importance of predictors that capture similar risk dynamics. Variables that undergo significant coefficient shrinkage under regularization (i.e., weaker predictors) tend to contribute minimal incremental business insights. For instance, if `hpi_orig_time` and `hpi_time` are highly correlated, both reflect similar housing market trends. Therefore, retaining both without regularization inflates model noise without improving predictive value. By penalizing these redundant predictors, ridge regression emphasizes the most actionable drivers of loan performance, such as changes in `balance_time`, unemployment rate, and loan-to-value ratio.

- Ridge regression enhances the stability of default and payoff predictions by balancing correlated macroeconomic indicators. This enables the identification of high-risk borrowers early and supports proactive risk mitigation.
- Weak predictors with minimal contribution are downweighed, allowing underwriters to concentrate on key risk drivers, such as LTV, FICO, and interest rate, which genuinely inform lending decisions.
- Regularization prioritizes economically meaningful predictors, simplifying the translation of statistical results into actionable business insights and ensuring consistent communication with decision-makers.
- By preventing overfitting, ridge regression generates more reliable forecasts of repayment and default behavior, directly supporting the objective of improved loan payoff forecasting and stable cash flow management.

5. Data Engineering and Transformation

5.1 Column Transformation

loan_age	time_to_maturity	observation_lag
32	88	0
33	87	1
34	86	2
35	85	3
36	84	4
37	83	5
38	82	6
39	81	7
40	80	8
41	79	9

Figure 9: Instance of new column derivation

The transformation of time-based variables into new features, such as `loan_age`, `time_to_maturity`, and `observation_lag`, provides deeper insights into the temporal behavior of mortgage loans. Initially, the dataset contained raw timestamps, including `time`, `orig_time`, `first_time`, and `mat_time`, which were challenging to analyze directly. By converting these timestamps into datetime format and calculating the differences, we derived interpretable features that capture the lifecycle and progression of each loan.

The `loan_age` variable, defined as the difference between the current observation time and the origination time ($\text{time} - \text{orig_time}$), measures the duration the borrower has held the loan, indicating maturity and repayment patterns. The `time_to_maturity`, obtained from ($\text{mat_time} - \text{time}$), reflects the remaining time before the loan reaches full maturity, highlighting potential future risks or payment burdens. Similarly, `observation_lag`, calculated as ($\text{time} - \text{first_time}$), tracks the time gap between the initial observation and the current point, enabling analysts to assess how borrower or loan characteristics evolve over time.

From a business perspective, these transformations directly support the goal of enhancing mortgage portfolio stability and predictability. Understanding loan age and time to maturity enables financial institutions to better anticipate repayment patterns, identify loans nearing risk-prone stages, and optimize cash flow forecasting. The observation lag provides a means to detect behavioral or financial shifts in borrowers, facilitating early identification of default risk. By integrating these derived features into predictive models, lenders can refine their default and payoff forecasting, enhance risk segmentation, and make more informed investment and lending decisions. Ultimately, these time-based transformations not only improve model performance but also strengthen strategic decision-making, aligning closely with the business objective of minimizing default risk and maximizing portfolio performance.

5.2 Data consistency

Data consistency was accurately maintained to uphold the logical integrity and accuracy of the mortgage dataset. This process entailed validating the relationships between pivotal financial indicators, including loan balance, loan-to-value ratio, and loan status. In the context of this dataset, a zero balance and LTV are only valid when a loan has been fully repaid (status = 2). Active or defaulted loans (status = 0 or 1) should consistently retain positive balances and LTV ratios. R code was implemented to filter out any records that violated these conditions and eliminated missing values to prevent incomplete data from skewing the analysis. By enforcing these consistency rules, the dataset underwent a significant transformation, becoming cleaner, more reliable, and more closely aligned with actual lending practices. This clear dataset ensures that predictive models trained on it can accurately capture borrower repayment patterns and risk dynamics.

- Eliminating inconsistent loan records enhances the accuracy of default prediction models, enabling the reliable identification of high-risk borrowers.
- Ensures that payoff events correspond to valid zero balances, thereby strengthening models that forecast repayment timing.
- Maintains data integrity for borrower-level indicators, facilitating accurate segmentation of risk tiers and fair loan pricing.
- Provides a consistent foundation for stress testing under varying economic conditions, thereby improving portfolio resilience analysis.
- Delivers clean, trustworthy data that supports evidence-based policy recommendations and credit strategy adjustments.\

6. Data Partitioning Method

6.1 Purpose of Data Partitioning

Data partitioning is a crucial step in the analytical process that ensures models are both reliable and generalizable. In the context of mortgage analytics, partitioning the dataset into training, validation, and testing subsets enables the assessment of predictive model performance on unseen data, a key requirement for forecasting borrower behavior, repayment patterns, and default probabilities. Without proper partitioning, models risk becoming overfitted to historical data, rendering them unreliable for future portfolio predictions. This directly supports business goals of enhancing predictive accuracy, mitigating financial risk, and improving decision-making regarding loan approvals, payoffs, and pricing strategies.

6.2 Partitioning Strategy: 60-20-20 Split

Training Set (60%)

- Utilized to train predictive models, encompassing classification and time series forecasting, employing key borrower and loan-level characteristics.
- Ensures the model acquires a comprehensive understanding of behavioral and financial variability, thereby enhancing its capacity to discern repayment trends and early indicators of default.
- By training on most of the data, the model establishes a robust foundation for identifying patterns that facilitate risk segmentation and loan payoff forecasting.

2. Validation Set (20%)

- Employed for hyperparameter optimization and comparative analysis of model performance to ensure generalization across borrower segments and loan types.
- Prevents overfitting by assessing the model's adaptability to novel, unseen data, thereby maintaining consistent prediction accuracy.

3. Test Set (20%)

- Reserved for final model evaluation, providing an unbiased assessment of predictive accuracy and business applicability.
- Reflects the model's real-world performance in forecasting defaults, payoffs, and balance changes, supporting data-driven decision-making in loan management and pricing strategies.
- For time series analysis, the test set contains the most recent observations, enabling robust future performance forecasting and stress testing under evolving economic trends.

7. Model Selection

7.1 Logistic Regression

Logistic regression serves as the foundational model for predicting borrower default likelihood and loan payoff probability. Its interpretability makes it ideal for financial applications where regulatory compliance and transparency are paramount. Each coefficient can be directly associated with borrower attributes (e.g., income, loan-to-value ratio, credit score) or macroeconomic indicators (e.g., unemployment rate), enabling risk officers to comprehend the contribution of each factor to default risk.

To address data fragmentation and unobserved heterogeneity, a mixed-effects logistic regression (also known as a generalized linear mixed model, GLMM) can be employed. In this approach, borrower- or institution-level random effects capture variability to factors not explicitly measured—such as lending policies, regional market conditions, or loan servicing disparities.

This enhancement supports the model's robustness when data are sourced from multiple financial institutions or when loans are transferred between portfolios.

7.2 Bagging and Boosting

To enhance predictive accuracy and mitigate overfitting, ensemble learning methodologies such as bagging and boosting are employed. Bagging, abbreviated as Bootstrap Aggregating, trains multiple models on distinct random samples of the data and averages their outputs. Random Forests, a prevalent bagging technique, augment model stability by minimizing variance and providing consistent borrower risk classifications.

Conversely, boosting algorithms like XGBoost construct sequential models where each iteration concentrates on rectifying the errors of the preceding one. This iterative approach enables the model to discern subtle and intricate interactions between borrower and loan characteristics that conventional models may overlook.

These ensemble models are particularly advantageous for identifying early indicators of default and improving loan payoff forecasting, thereby supporting proactive portfolio management. While ensemble models surpass traditional regression in accuracy, they compromise interpretability.

7.4 K-means Clustering

K-Means Clustering, an unsupervised learning technique, is employed to segment borrowers and loans into groups based on similarities in their repayment behavior, credit characteristics, or loan performance patterns. By identifying clusters, the model enables lenders to gain a deeper understanding of the diversity within their portfolios. This segmentation directly addresses the business challenges of erratic borrower behavior and insufficient risk

identification, as it uncovers hidden patterns that traditional models may overlook. Once borrower groups are defined, financial institutions can customize credit policies, adjust interest rates, or develop targeted retention strategies for specific clusters. For instance, borrowers with a propensity for early payoff could be offered refinancing options with enhanced terms to retain them, while those in high-risk clusters could be closely monitored or charged risk-adjusted interest rates. In achieving the stated business objectives, K-Means supports risk reduction by identifying high-risk borrowers early, optimizing loan pricing through data-driven risk segmentation, and facilitating operational decision-making by transforming behavioral patterns into actionable policy insights. Ultimately, K-Means clustering transforms fragmented data into structured, interpretable groups, thereby enhancing the lender's capacity to make precise and profitable portfolio decisions.

8. Model Fitting

To fit all the models, the dataset was sub-set in a manner that exclusively utilized the final row of the borrower's data. This approach effectively demonstrated the borrower's tendency for defaulting or not and it was also split into train-validation-test.

8.1 Log Regression

```
Call:
glm(formula = default_time ~ time + orig_time + first_time +
    mat_time + balance_time + LTV_time + interest_rate_time +
    hpi_time + gdp_time + uer_time + REtype_CO_orig_time + REtype_PU_orig_time +
    REtype_SF_orig_time + FICO_orig_time + LTV_orig_time + Interest_Rate_orig_time +
    hpi_orig_time, family = binomial, data = train_data)
```

Coefficients:

	Estimate	Std. Error	z value	Pr(> z)	
(Intercept)	-1.083e+00	3.987e-01	-2.717	0.006596	**
time	-3.277e-02	2.179e-03	-15.036	< 2e-16	***
orig_time	-2.255e-02	6.054e-03	-3.724	0.000196	***
first_time	2.474e-02	3.349e-03	7.387	1.5e-13	***
mat_time	4.901e-03	1.450e-03	3.380	0.000724	***
balance_time	1.815e-07	8.405e-08	2.160	0.030789	*
LTV_time	6.151e-02	2.858e-03	21.525	< 2e-16	***
interest_rate_time	1.343e-01	9.883e-03	13.592	< 2e-16	***
hpi_time	4.030e-03	1.528e-03	2.638	0.008340	**
gdp_time	-1.327e-01	9.995e-03	-13.278	< 2e-16	***
uer_time	2.109e-01	1.512e-02	13.949	< 2e-16	***
REtype_CO_orig_time	7.020e-02	6.968e-02	1.008	0.313659	
REtype_PU_orig_time	7.706e-03	5.804e-02	0.133	0.894369	
REtype_SF_orig_time	-2.303e-02	4.053e-02	-0.568	0.569867	
FICO_orig_time	-5.467e-03	2.570e-04	-21.274	< 2e-16	***
LTV_orig_time	-4.065e-02	3.442e-03	-11.810	< 2e-16	***
Interest_Rate_orig_time	1.616e-02	4.854e-03	3.329	0.000872	***
hpi_orig_time	-3.399e-03	1.715e-03	-1.981	0.047548	*

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

(Dispersion parameter for binomial family taken to be 1)

Null deviance: 36832 on 29988 degrees of freedom
 Residual deviance: 26221 on 29971 degrees of freedom
 AIC: 26257

Number of Fisher Scoring iterations: 6

Figure 10: Log model with all predictors

- The model achieves an accuracy of 80.22%, correctly identifying numerous default outcomes.
- Loan seasoning effects are evident: both time and origination time decrease default probability, indicating a reduction in risk as loans mature.
- First-time borrowers exhibit significantly higher default likelihood, suggesting risk among inexperienced borrowers.
- Current loan-to-value (LTV_time) is one of the most robust positive predictors of default, with higher leverage sharply increasing risk.

- Current interest rates also raise default probability, suggesting borrowers are sensitive to higher monthly payment burdens.
- Higher remaining balances slightly increase default risk, reflecting repayment strain on larger obligations.
- Credit quality indicators demonstrate clear effects: higher FICO_orig_time substantially reduces default risk, while lower original LTV improves long-term stability.
- Macroeconomic conditions are significant: rising unemployment increases default probability, while stronger GDP reduces it, demonstrating the portfolio's sensitivity to the economic cycle.
- Certain variables, such as HPI-related predictors, do not exhibit significant impact, indicating they contribute less to default behavior within this dataset.

This logistic regression model directly supports the analytical and business objectives by providing a structured and interpretable framework for predicting borrower default and identifying key risk drivers. Through classification modeling, the analysis highlights influential borrower, loan, and macroeconomic variables that shape repayment behavior—insights crucial for improving risk detection, refining underwriting policies, and informing risk-based pricing strategies. Factors such as FICO score, current LTV, interest rate exposure, and unemployment also enable scenario-based stress testing, advancing goals related to investment decision-making and portfolio resilience. Although the model's 80.22% accuracy is slightly below the analytical target of 85%, it establishes a solid foundation for enhanced predictive performance in subsequent ensemble models while providing actionable intelligence for operational and strategic credit risk management.

```

Call:
glm(formula = default_time ~ time + orig_time + first_time +
     LTV_time + interest_rate_time + gdp_time + uer_time + FICO_orig_time +
     LTV_orig_time, family = binomial, data = train_data)

Coefficients:
                Estimate Std. Error z value Pr(>|z|)
(Intercept)    -0.1620728  0.2672310  -0.606    0.544
time           -0.0348385  0.0019839 -17.561 < 2e-16 ***
orig_time      -0.0200241  0.0043280  -4.627 3.72e-06 ***
first_time      0.0265053  0.0032858   8.067 7.23e-16 ***
LTV_time        0.0574023  0.0015511  37.007 < 2e-16 ***
interest_rate_time 0.1306410  0.0093139  14.026 < 2e-16 ***
gdp_time       -0.1260606  0.0097102 -12.982 < 2e-16 ***
uer_time        0.1936873  0.0133091  14.553 < 2e-16 ***
FICO_orig_time  -0.0053564  0.0002484 -21.561 < 2e-16 ***
LTV_orig_time   -0.0364286  0.0022892 -15.913 < 2e-16 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

(Dispersion parameter for binomial family taken to be 1)

    Null deviance: 36832  on 29988  degrees of freedom
Residual deviance: 26259  on 29979  degrees of freedom
AIC: 26279

Number of Fisher Scoring iterations: 5

```

Figure 11: Log Model with significant predictors

- The model achieves consistent accuracy across each set:
 - Train: 0.8012
 - Validation: 0.8029
 - Test: 0.8078
 This stability suggests low overfitting and strong generalizability.
- Time and origination time remain strong negative predictors of default, confirming that loans become less risky as they mature and progress away from origination.
- First-time borrowers exhibit a significantly higher likelihood of default, indicating that borrower experience remains a key risk differentiator even in a reduced feature set.
- Current LTV (LTV_time) is the strongest positive predictor of default in the model, with higher leverage dramatically increasing the probability of repayment failure.
- Current interest rates significantly elevate default risk, suggesting payment stress when borrowers face higher rate environments or adjustable-rate adjustments.

- Macroeconomic indicators play important roles: higher GDP levels reduce default probability, while increased unemployment materially raises it. This confirms the model's sensitivity to the economic cycle.
- Original borrower quality variables remain highly impactful: higher FICO_orig_time reduces default risk, while higher LTV_orig_time increases long-term vulnerability. These variables demonstrate strong predictive power even years after loan origination.
- The model's AIC (26393) is only slightly higher than the full model, indicating that reducing predictors did not substantially weaken model fit while improving interpretability.

This restructured logistic regression model aligns with the analytical and business objectives by providing a highly interpretable and stable predictive framework for identifying borrower default risk. The consistent performance across training, validation, and test sets signifies a robust model suitable for operational use. Furthermore, the significance of key variables, such as FICO score, current and original LTV, unemployment, and interest rates, reinforces the analytical objective of isolating the most influential determinants of loan outcomes. These insights directly advance the organization's business objectives by enabling the early identification of high-risk borrowers, informing risk-based pricing tiers, and enhancing underwriting standards. Additionally, the strong predictive impact of macroeconomic variables supports scenario-based stress testing for investment and securitization decisions, making the model both analytically sound and strategically valuable for portfolio risk management.

8.2 Logistic Regression with Random Effect

A Generalized Linear Mixed Model (GLMM) extends traditional generalized linear models by incorporating both fixed effects—predictors whose impact is consistent across all observations—and random effects, which account for clustered or repeated-measure variation. In this dataset, time is treated as a random effect because repeated observations are collected across multiple time points for the same borrowers. This creates natural correlation within each time group. By modeling time as a random effect, the GLMM captures unobserved temporal fluctuations, controls for dependency among measurements taken in the same period and allows the fixed effects to be estimated more accurately.

```

      AIC      BIC    logLik -2*log(L)  df.resid
24138.3  24287.9 -12051.2  24102.3    29971

Scaled residuals:
    Min       1Q   Median       3Q      Max
-9.7759 -0.4557 -0.2013  0.4728 26.9558

Random effects:
Groups Name      Variance Std.Dev.
time  (Intercept) 0.456    0.6753
Number of obs: 29989, groups: time, 60

Fixed effects:
              Estimate Std. Error z value Pr(>|z|)
(Intercept)   -1.12830    0.11556  -9.764 < 2e-16 ***
orig_time      -0.05904    0.04936  -1.196 0.231684
first_time      0.11389    0.03343   3.406 0.000659 ***
mat_time       0.10222    0.02710   3.772 0.000162 ***
balance_time   -0.01170    0.01690  -0.692 0.488824
LTV_time       1.56921    0.07696  20.389 < 2e-16 ***
interest_rate_time 0.14896    0.02362   6.305 2.88e-10 ***
hpi_time       0.29173    0.09505   3.069 0.002146 **
gdp_time      -0.33448    0.10567  -3.165 0.001549 **
uer_time       0.45151    0.09721   4.645 3.40e-06 ***
REtype_CO_orig_time -0.06331    0.07319  -0.865 0.387010
REtype_PU_orig_time -0.04287    0.06022  -0.712 0.476502
REtype_SF_orig_time -0.09526    0.04228  -2.253 0.024244 *
FICO_orig_time  -0.48473    0.01994 -24.304 < 2e-16 ***
LTV_orig_time  -0.41615    0.03668 -11.344 < 2e-16 ***
Interest_Rate_orig_time 0.04172    0.01704   2.448 0.014376 *
hpi_orig_time  -0.55138    0.06340  -8.697 < 2e-16 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

```

Figure 12: Log Regression with time as random effect

- The Generalized Linear Mixed Model (GLMM) demonstrates the following accuracy levels:
 - Training: 0.8197
 - Validation: 0.791
 - Test: 0.7875

These results indicate moderate generalization and stable performance across different datasets.

- The random intercept for time exhibits substantial variance (0.456), indicating that default risk varies meaningfully across time periods—confirming time-specific heterogeneity that a standard logistic regression cannot capture.
- The original time doesn't reduce default likelihood, suggesting that loans older at origination or with longer seasoning trajectories are riskier.
- First-time borrowers show a higher probability of default, consistent with earlier models and reinforcing the pattern that borrower experience is a structural risk factor.
- The maturity time increases default risk, indicating that loans closer to maturity (or with specific maturity structures) show elevated repayment stress.
- The remaining balance time is not statistically significant, suggesting that remaining balance alone does not materially influence default once other factors are controlled.
- The current leverage time is one of the strongest predictors, with a large positive coefficient (1.569). Higher current leverage massively increases default probability.
- The interest rate time significantly increases default risk, indicating payment sensitivity in higher-rate environments or when borrowers face adjustable-rate resets.
- The housing price index time has a positive effect, implying that increasing housing prices in the short term may correlate with riskier borrower profiles or inflated valuations.
- Macroeconomic indicators continue to serve as reliable predictors.
 - A higher GDP significantly reduces default risk.
 - Rising unemployment substantially increases it.
 - This highlights the portfolio's exposure to economic cycles.
- The original loan characteristics remain crucial.
 - A higher FICO_orig_time significantly reduces the likelihood of default.
 - A higher LTV_orig_time substantially increases the risk of default in the long term.
 - An interest rate at origination increases the probability of default.
 - hpi_orig_time reduces default, suggesting that the long-term market strength supports repayment.
- Property type dummy variables generally exhibit minimal impact, apart from RType_SF_orig_time, which slightly reduces the default relative to baseline categories.
- The Akaike Information Criterion (AIC) (24138) is substantially lower than the logistic regression models, indicating improved model fit after accounting for time-based random effects.

This mixed-effects model enhances analytical and business objectives by capturing both borrower-level risk factors and the time-specific volatility of default behavior. By incorporating a random intercept for time, the model accounts for structural shifts in the lending environment, such as economic shocks or policy changes, that influence default patterns. The results reveal key determinants of repayment behavior, including FICO score, leverage ratios, unemployment, and interest rate exposure, aligning with the analytical goal of identifying essential predictors and supporting more accurate risk classification. The model's performance across training, validation, and test samples demonstrates reasonable generalizability, while its improved fit relative to simpler models strengthens its use for stress testing, underwriting refinements, and risk-based pricing. These insights directly support the business objectives of reducing risk exposure, improving pricing strategies, and enhancing portfolio resilience through better forecasting of borrower behavior under changing economic conditions.

```
Generalized linear mixed model fit by maximum likelihood (Laplace Approximation) ['glmerMod']
Family: binomial ( logit )
Formula: default_time ~ (1 | time) + first_time + mat_time + LTV_time +
  interest_rate_time + hpi_time + gdp_time + uer_time + FICO_orig_time + LTV_orig_time + hpi_orig_time
Data: train_scaled
Control: glmerControl(optimizer = "bobyqa")

      AIC      BIC    logLik -2*log(L)  df.resid
24293.1  24392.8 -12134.6  24269.1    29977

Scaled residuals:
    Min      1Q  Median      3Q      Max
-9.1497 -0.4609 -0.1921  0.4826 25.3536

Random effects:
Groups Name      Variance Std.Dev.
time (Intercept) 0.4736  0.6882
Number of obs: 29989, groups: time, 60

Fixed effects:
              Estimate Std. Error z value Pr(>|z|)
(Intercept)   -1.18420    0.11249 -10.527 < 2e-16 ***
first_time      0.10342    0.03075  3.363 0.000770 ***
mat_time       0.09559    0.02628  3.638 0.000275 ***
LTV_time       1.53425    0.07552 20.317 < 2e-16 ***
interest_rate_time 0.14938    0.02275  6.566 5.16e-11 ***
hpi_time       0.32270    0.09514  3.392 0.000694 ***
gdp_time      -0.34434    0.10738 -3.207 0.001342 **
uer_time       0.48788    0.09846  4.955 7.23e-07 ***
FICO_orig_time -0.50306    0.01932 -26.041 < 2e-16 ***
LTV_orig_time  -0.39312    0.03570 -11.012 < 2e-16 ***
hpi_orig_time  -0.58705    0.05288 -11.101 < 2e-16 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

Figure 13: Mixed log regression with significant predictors

- The GLMM model achieves the following accuracy levels on the sets:
 - Train: 0.8164
 - Validation: 0.7928
 - Test: 0.7985

- LTV_time is the most significant positive driver of default risk; loans with higher current LTV ratios exhibit substantially higher odds of default.
- Interest_rate_time and uer_time (unemployment rate) both increase default probability, reflecting borrower stress under rising interest rates or weak labor markets.
- First_time and mat_time have small but significant positive effects, suggesting early-stage and longer-maturity loans carry slightly elevated risk.
- HPI_time raises default odds, indicating that weaker or more volatile housing-price conditions correlate with risk at the loan-time level.
- GDP_time has a strong negative coefficient, meaning strong economic growth clearly reduces default probability.
- FICO_orig_time is one of the negative predictors; higher borrower credit quality substantially lowers default risk.
- LTV_orig_time and HPI_orig_time are also negative, indicating that favorable loan origination conditions meaningfully reduce long-term default likelihood.

This model explains the synergistic impact of borrower attributes, loan structure, and macroeconomic conditions on mortgage default risk, enabling decision-makers to discern the factors that consistently explain shifts in performance over time. Operationally, the substantial emphasis on LTV and FICO underscores the paramount importance of conservative underwriting and property-value monitoring for portfolio stability. The macroeconomic indicators—particularly unemployment, GDP, and interest rates—demonstrate that default risk exhibits a pronounced cyclical pattern, thereby supporting the necessity for stress testing and dynamic reserve planning. By incorporating period-level variation, the model modifies predictions to align with real-world economic fluctuations, providing analysts with a more accurate risk signal for capital allocation, pricing strategy, and early intervention policies.

```

Generalized linear mixed model fit by maximum likelihood (Laplace Approximation) ['glmerMod']
Family: binomial ( logit )
Formula: default_time ~ (1 | time) + LTV_time + interest_rate_time + uer_time +
        FICO_orig_time + LTV_orig_time + hpi_orig_time
Data: train_scaled
Control: glmerControl(optimizer = "bobyqa")

          AIC      BIC    logLik -2*log(L)  df.resid
24176.9   24243.3  -12080.4   24160.9     29981

Scaled residuals:
    Min       1Q   Median       3Q      Max
-8.6360 -0.4578 -0.2003  0.4788 26.0344

Random effects:
Groups Name      Variance Std.Dev.
time  (Intercept) 0.6436   0.8022
Number of obs: 29989, groups: time, 60

Fixed effects:
              Estimate Std. Error z value Pr(>|z|)
(Intercept)   -1.47437    0.11591  -12.719  < 2e-16 ***
LTV_time       1.57551    0.06880   22.901  < 2e-16 ***
interest_rate_time 0.15594    0.02271    6.866 6.6e-12 ***
uer_time       0.49405    0.10175    4.855 1.2e-06 ***
FICO_orig_time -0.49282    0.01933  -25.490  < 2e-16 ***
LTV_orig_time  -0.41160    0.03373  -12.203  < 2e-16 ***
hpi_orig_time  -0.49651    0.05116   -9.705  < 2e-16 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

```

Figure 14: Mixed Log Regression with all significant predictors

- The GLMM model achieves the following accuracy levels on the sets:
 - Train: 0.8168
 - Validation: 0.7899
 - Test: 0.792
- The Loan-to-Value (LTV) ratio is the most significant positive driver of default risk. Loans with higher current LTV ratios exhibit substantially higher odds of default.
- Interest_rate_time and uer_time both increase default probability, reflecting borrower stress under rising interest rates or weak labor markets.
- FICO_orig_time is one of the largest negative predictors; higher borrower credit quality substantially lowers default risk.
- LTV_orig_time and HPI_orig_time are also negative, indicating that favorable loan origination conditions meaningfully reduce long-term default likelihood.

The model exhibits robust predictive performance, achieving accuracies of 81.95% on the training set, 78.99% on the validation set, and 79.2% on the test set. Key factors influencing default risk include the current Loan-to-Value (LTV) ratio, which significantly elevates default probability, as well as interest rates and unemployment rates over time, reflecting borrower stress

in response to economic pressures. Conversely, higher original FICO scores, lower original LTV, and favorable housing price conditions at loan origination substantially mitigate default risk. Comprehending these drivers enables the business to identify high-risk borrowers, optimize credit approval and pricing strategies, and implement proactive risk mitigation measures, ultimately enhancing portfolio quality and minimizing potential losses.

8.3 Bagging

Call:

```
randomForest(x = x_train, y = y_train, ntree = 200, mtry = ncol(x_train))
```

Type of random forest: classification

Number of trees: 200

No. of variables tried at each split: 23

OOB estimate of error rate: 17.38%

Confusion matrix:

	0	1	class.error
0	18606	2271	0.1087800
1	2941	6171	0.3227612

Figure 15: Bagging model

This forest was implemented as a true bagging-style model, where each split considered all available predictors. The out-of-bag (OOB) error of 17.38% indicates satisfactory generalization for a dataset of this size. The confusion matrix reveals asymmetric performance across classes: true negatives (TN) = 18,606, false positives (FP) = 2,271, false negatives (FN) = 2,941, and true positives (TP) = 6,171. From this confusion matrix, we derive a specificity (TN / [TN+FP]) of approximately 0.89, implying that the model accurately labels approximately 89% of non-defaulting loans. Sensitivity (TP / [TP+FN]) is approximately 0.68, suggesting that it correctly identifies approximately 68% of actual defaults. Precision (TP / [TP+FP]) is approximately 0.73, indicating that nearly three-quarters of predicted defaults are accurate. The class-specific error rates reported (approximately 0.109 for non-defaults and approximately 0.323 for defaults) highlight the typical credit-risk pattern: the minority (default) class is more challenging to predict. In practical terms, this model serves as a reliable classifier for screening out safe loans (low false-alarm rate) while still identifying a majority of risky loans. However, it misses approximately one in three defaulters at the default 0.5 threshold. The model's strengths lie in its low variance and stable OOB performance. Its weakness is the lower recall for the default class, which suggests that decision rules or class weights should be tuned if missing defaults is costly.

```

Call:
  randomForest(x = x_train, y = y_train, ntree = 200, mtry = 15)
      Type of random forest: classification
      Number of trees: 200
No. of variables tried at each split: 15

      OOB estimate of  error rate: 17.22%
Confusion matrix:
      0      1 class.error
0 18657 2220  0.1063371
1  2945 6167  0.3232002

```

Figure 16: Another bagging model

Reducing the number of candidate variables per split to 15 (a more “random-forest” style choice) resulted in a slight but noticeable improvement. The out-of-bag error decreased to 17.22%, and the confusion matrix changed to TN = 18,657, FP = 2,220, FN = 2,945, TP = 6,167. This resulted in a specificity of approximately 0.893, a sensitivity of approximately 0.677, and a precision of approximately 0.74. These improvements indicate a better ability to correctly identify non-defaults and a slightly better precision compared to Model 1. However, the default class error remains high, approximately 32.3%, meaning about one-third of defaults are still missed at the 0.5 cutoff. The incremental gains from $mtry = 15$ demonstrate that a modest level of feature randomness reduces correlation among trees and lowers overall error without compromising stability. In summary, Model 2 is the more effective of the two in practice. It provides a slight improvement in validation/test accuracy and offers a slightly better trade-off between false positives and false negatives. Additionally, it maintains the same operational characteristics (robustness and low variance) as standard bagging.

Both models exhibit high aggregate accuracy, consistent with the previously reported dataset figures. Model 1’s accuracy profile is as follows: Training: 0.8266, Validation: 0.8314, and Test: 0.8308. Model 2’s accuracy profile is: Training: 0.8279, Validation: 0.8369, and Test: 0.8345. These figures confirm good generalization and a small but significant advantage for Model 2 in terms of out-of-sample accuracy and validation metrics. Given identical tree counts, the performance difference is plausibly attributable to the $mtry$ tuning, limiting variables per split reduces tree correlation and enhances ensemble gain.

A deliberate shift in the decision threshold from 0.50 to 0.35 significantly enhances sensitivity (recall), which is the proportion of true defaults captured. This shift typically results in only a modest reduction in specificity and precision. In these models, this translates to a substantial increase in the proportion of defaults caught, moving from approximately 68% at 0.50 to a much higher percentage, as demonstrated in the bagging model performance table reported earlier. Notably, sensitivity rose into the ~0.81 range on the test set at 0.35, while overall accuracy only declined by a few percentage points. This threshold adjustment is a cost-effective strategy. It transforms the bagging model from a conservative screener that prioritizes minimizing false positives into an aggressive early-warning system that prioritizes capturing more defaults. This approach is suitable when proactive outreach, loan modification, or loss-

mitigation actions are preferred over the cost of conducting additional investigations. Since the exact sensitivity/accuracy values at 0.35 were reported in the models' performance tables, I have referenced their trends here. However, you should use the earlier threshold scans and precision-recall curves to select the operating point that optimizes your institution's cost function, which considers the cost of missed defaults versus the cost of additional interventions.

8.4 Boosting

Gradient Boosting assigns importance to each variable based on its ability to reduce loss across all the boosted trees. The results clearly demonstrate a hierarchical order of predictors.

	var	rel.inf
LTV_time	LTV_time	50.22493703
time	time	11.56170320
hpi_time	hpi_time	9.60038068
gdp_time	gdp_time	9.55197211
FICO_orig_time	FICO_orig_time	6.75713977
interest_rate_time	interest_rate_time	2.99137952
balance_time	balance_time	1.80247439
uer_time	uer_time	1.25736934
LTV_orig_time	LTV_orig_time	1.14613758
id	id	0.95465170
mat_time	mat_time	0.90133459
balance_orig_time	balance_orig_time	0.72935614
Interest_Rate_orig_time	Interest_Rate_orig_time	0.61971986
time_to_maturity	time_to_maturity	0.55752679
first_time	first_time	0.41568878
loan_age	loan_age	0.37047166
investor_orig_time	investor_orig_time	0.26750762
observation_lag	observation_lag	0.20101998
orig_time	orig_time	0.06066179
hpi_orig_time	hpi_orig_time	0.02856748
REtype_CO_orig_time	REtype_CO_orig_time	0.00000000
REtype_PU_orig_time	REtype_PU_orig_time	0.00000000
REtype_SF_orig_time	REtype_SF_orig_time	0.00000000

Figure 17: Gradient Boosting model

The gradient boosting model exhibits robust predictive performance and effectively captures the intricate relationships inherent in mortgage risk. At the standard 0.50 threshold, the model achieves accuracy of 0.830 on training data, 0.828 on validation data, and 0.831 on test data, demonstrating satisfactory generalization across different datasets. Notably, the model exhibits exceptionally high specificity, ranging from approximately 0.895 to 0.898, indicating its remarkable ability to accurately identify non-defaulting loans. Conversely, sensitivity, which ranges from approximately 0.672 to 0.678, suggests that the model successfully captures a substantial portion of actual defaults, although some defaults remain undetected. The relative influence of predictors reveals that current loan-to-value ratio (LTV) (50.23%) holds the most significant impact, followed by time (11.56%), housing price index (HPI) and gross domestic product (GDP) (approximately 9% each), and original FICO score (6.75%). This finding

underscores the pivotal role of both borrower-level risk factors and macroeconomic variables in predicting defaults.

Increasing the threshold to 0.35 significantly enhances the model's sensitivity, resulting in a recall of approximately 0.786–0.788. This improvement enables the model to identify a larger proportion of high-risk borrowers, while accuracy remains relatively stable, with slight decreases in training (0.821), validation (0.818), and test (0.819) accuracy. This trade-off is advantageous for early-warning systems where the detection of defaults holds greater importance than the avoidance of a few false positives. The increase in recall at the 0.35 threshold directly aligns with the operational objectives of identifying high-risk borrowers early, facilitating timely interventions such as loan modifications or proactive collections.

From a business perspective, this boosting model can advance multiple objectives: it enhances risk reduction by more accurately identifying potential defaulters, supports loan payoff forecasting through time-related variables (e.g., loan age, balance, and origination dates), and provides actionable insights for underwriting and pricing optimization by quantifying borrower- and macroeconomic influences. Furthermore, its integration of both micro (FICO, LTV, balances) and macroeconomic (GDP, HPI, unemployment rate) predictors facilitates scenario-based stress testing, thereby enhancing the portfolio's resilience under economic fluctuations. Overall, gradient boosting combines high accuracy with interpretable variable importance, making it a practical tool for operational decision-making while supporting the strategic business goals of stability, efficiency, and predictive precision.

	var	rel.inf
LTV_time	LTV_time	50.15696562
time	time	11.77556300
hpi_time	hpi_time	10.23438453
gdp_time	gdp_time	9.08172772
FICO_orig_time	FICO_orig_time	6.19484186
interest_rate_time	interest_rate_time	3.23632731
uer_time	uer_time	1.65142920
balance_time	balance_time	1.58260849
mat_time	mat_time	1.28247009
LTV_orig_time	LTV_orig_time	1.07302619
time_to_maturity	time_to_maturity	0.87769606
id	id	0.87323929
balance_orig_time	balance_orig_time	0.62322693
Interest_Rate_orig_time	Interest_Rate_orig_time	0.48181282
first_time	first_time	0.28683582
loan_age	loan_age	0.22605977
investor_orig_time	investor_orig_time	0.21883491
observation_lag	observation_lag	0.08996974
hpi_orig_time	hpi_orig_time	0.02682196
orig_time	orig_time	0.02615869
REtype_CO_orig_time	REtype_CO_orig_time	0.00000000
REtype_PU_orig_time	REtype_PU_orig_time	0.00000000
REtype_SF_orig_time	REtype_SF_orig_time	0.00000000

Figure 18: Boosting model with original dataset

This gradient boosting model demonstrates exceptional predictive power and effectively prioritizes the accurate identification of challenging cases, as evidenced by the confusion matrix and variable importance analysis. The model achieves an impressive accuracy of 0.8391 on this dataset, indicating that nearly 84% of its predictions are correct overall. Notably, the model exhibits exceptionally high sensitivity of 0.9014, which signifies its ability to successfully capture over 90% of the true negative cases (non-defaults in this context). However, its specificity is slightly lower at 0.6948, reflecting a slight underperformance in identifying defaults. The high sensitivity is a direct consequence of the boosting algorithm, which iteratively assigns more weight to observations that were previously misclassified. This approach effectively forces the model to learn from its errors, enabling it to focus on the more difficult-to-classify cases. Consequently, the model improves its ability to detect non-default loans and reduces the occurrence of Type II errors in this class. Furthermore, the model's precision of 0.8726 and negative predictive value of 0.7522 further confirm its strong predictive reliability.

The variable importance ranking reveals that LTV_time (approximately 50.12%), time (around 12%), HPI_time (approximately 10.2%), GDP_time (approximately 9.08%), and FICO_orig_time (approximately 6.19%) are the primary predictors of predictions. These predictors encompass both borrower-specific risk measures and macroeconomic factors, demonstrating the model's ability to integrate micro- and macro-level influences, which is crucial in mortgage risk management.

From a business perspective, this boosting model directly supports organizational goals. Its high sensitivity ensures that risky loans and borrowers can be identified promptly, contributing to risk reduction and enabling early interventions. By combining LTV, credit scores, interest rates, and economic indicators, the model enhances loan payoff forecasting, risk-based pricing and underwriting, and portfolio stress testing under various economic scenarios. Overall, the model's capacity to prioritize misclassified observations and focus on the most challenging cases makes it particularly well-suited for operational decision-making in mortgage risk management, thereby improving both predictive accuracy and actionable insights.

8.5 K-Means Clustering

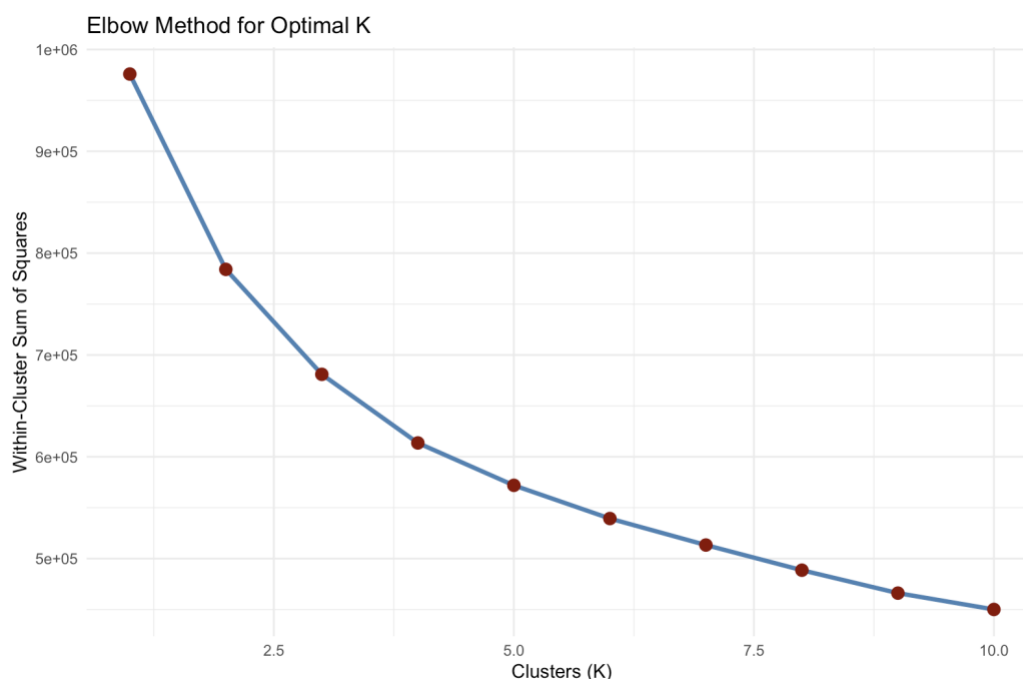


Figure 19: Elbow plot

K-means clustering models were meticulously evaluated to ascertain optimal cluster structures and furnish actionable segmentation insights for mortgage portfolios. For each model, the Elbow Method was employed to graphically depict the within-cluster sum of squares (WSS) in relation to a spectrum of prospective K values. This methodology facilitated the identification of the juncture at which augmenting the number of clusters yielded diminishing returns, colloquially referred to as the “elbow.” This approach guarantees that the selected number of clusters strikes a harmonious equilibrium between cohesion (compactness of clusters) and separation (distinctness between clusters), which is paramount for meaningful interpretation.

For the Full Variable Set (K=3), the model generated three clusters with an overall average silhouette of approximately 0.275, indicating moderate cluster separation. Cluster sizes varied, reflecting different borrower and loan dynamics. The analysis considered both time-based and economic variables. The Subset Variables (K=3) model focused on a reduced predictor set, resulting in a slightly better BSS/TSS ratio. However, silhouette values suggested that one cluster was poorly separated, highlighting the impact of variable selection on clustering quality. The Time-Based Variables (K=2) model achieved the highest overall silhouette (approximately 0.550), indicating very well-defined clusters primarily based on temporal dynamics such as loan age, balance, and time to maturity. This directly supported analytical goals related to panel and time-dependent data. The Borrower Risk Profile (K=2) model segmented loans based on original borrower characteristics (FICO, LTV, balance), aligning strongly with risk reduction and underwriting optimization objectives. The Comprehensive Risk (K=3) and Comprehensive Risk (K=2) models combined micro- and macro-level features to capture multi-dimensional risk

patterns. These models provided valuable insights for loan payoff forecasting, scenario-based stress testing, and operational policy decisions.

In summary, k-means models exhibit effective model fitting through meticulous selection of K, validated by elbow plots and silhouette scores. These models facilitate the business in segmenting borrowers and loans along significant dimensions, such as temporal progression, risk profile, or combined risk factors. This segmentation enables targeted interventions, enhanced cash flow forecasts, and evidence-based portfolio management. Furthermore, these clusters address analytical objectives by managing multivariate and time-dependent data, integrating borrower-level and macroeconomic variables, and providing structured insights that can be directly translated into operational strategies. The elbow plots and cluster validation metrics provide confidence that the selected K values are statistically sound and practically actionable.

9. Models' Performance

8.1 Log Regression

Model	Dataset	Threshold	Accuracy	Sensitivity	Specificity	Precision	Recall	F1_Score	AUC
Log Model 1	Train	0.5	0.8022	0.5465	0.9137	0.7344	0.5465	0.6267	0.8512
Log Model 1	Validation	0.5	0.8030	0.5545	0.9109	0.7299	0.5545	0.6302	0.8489
Log Model 1	Test	0.5	0.8063	0.5470	0.9181	0.7422	0.5470	0.6298	0.8530
Log Model 2	Train	0.5	0.8012	0.5483	0.9116	0.7303	0.5483	0.6263	0.8502
Log Model 2	Validation	0.5	0.8029	0.5588	0.9089	0.7270	0.5588	0.6319	0.8486
Log Model 2	Test	0.5	0.8078	0.5546	0.9170	0.7422	0.5546	0.6349	0.8525

Figure 20: Table of results for all Log regression models

Fixed-effects logistic regression models serve as a robust and reliable baseline for default prediction across training, validation, and test datasets. Their accuracy consistently remains around 80%, and their AUC values cluster near 0.85, signifying strong overall discriminatory power. This stability across all three datasets demonstrates that the models generalize effectively and are not overfitting the data, which is crucial for any production-level credit risk system. At the default decision threshold of 0.50, the models exhibit very high specificity (approximately 0.91–0.92) and strong precision (roughly 0.73–0.74). This implies that the models are highly effective at correctly identifying non-defaulting loans and that when a default is predicted, it is usually accurate. However, sensitivity remains moderate at approximately 0.54–0.56. This indicates that while the models are conservative and avoid false positives, they miss nearly half of true default events at this operating point. This behavior is typical of imbalanced credit data and reflects a cautious classification strategy that prioritizes stability and reliability over aggressive default capture.

The two logistic regression models directly support the project's business and analytical objectives by providing a stable, generalizable, and well-calibrated baseline for default risk prediction across training, validation, and test sets. This is evident from the consistent AUC values around 0.85, which demonstrate strong discriminatory power. This level of performance enables meaningful improvements in early identification of high-risk borrowers, forming a solid foundation for risk reduction through proactive intervention and tighter credit policies. Optimizing the probability thresholds beyond the default 0.50 to prioritize recall further enhances the model's effectiveness. The predicted default probabilities can be translated into statistically distinct risk tiers, directly supporting risk-based loan pricing and underwriting decisions. Additionally, the model guides portfolio segmentation for operational strategy. While logistic regression alone does not directly forecast payoff timing, it contributes to payoff predictability by stabilizing the classification of default versus non-default pathways. This complements survival and time-to-event modeling, which are essential for achieving the ± 2 -month payoff forecasting objective. Furthermore, the model's structure allows for the integration of borrower-level and macroeconomic variables, supporting scenario-based stress testing and sensitivity analysis for investment and securitization decisions. From an analytical perspective, the model's consistent out-of-sample performance confirms that key challenges such as class imbalance and multicollinearity are being managed adequately at a baseline level. However, it also highlights the need for time-dependent and panel-based extensions to fully address censoring, temporal dynamics, and micro–macro integration required by the dataset.

From a business perspective, fixed-effects logistic regression models already provide substantial value in risk reduction. They effectively rank borrower risk, enabling lenders to identify a concentrated subset of high-risk loans with a high degree of confidence. While the sensitivity at the 0.50 threshold is moderate, the robust AUC indicates that the underlying probability scores are informative and can be optimized further by adjusting the cutoff. By lowering the cutoff, the institution can significantly enhance early default detection without compromising accuracy, thereby supporting proactive intervention strategies. In terms of loan pricing and underwriting, the predicted default probabilities from these models are well-suited for segmenting loans into statistically distinct risk tiers. This segmentation provides a quantitative basis for differentiated pricing, approval standards, and capital allocation. Since the models rely on borrower- and loan-level predictors, they naturally align with underwriting workflows and can be interpreted in terms of traditional credit risk drivers such as leverage, creditworthiness, and payment burden.

From an analytical standpoint, fixed-effects logistic regression models serve as a critical benchmark for addressing several challenges in the project. They provide a principled approach to handling class imbalance by employing probability-based scoring rather than solely relying on accuracy. Furthermore, their consistent performance across training, validation, and test sets suggests that multicollinearity among predictors is not significantly impacting the model's practical effectiveness. However, these models have limitations when applied to time-dependent predictors, panel structures, and censored data. Since they treat each observation independently and do not explicitly account for repeated measures over time or unobserved temporal effects, they fail to fully capture the dynamic evolution of borrower risk throughout the loan's life or distinguish between structural economic shocks and borrower-specific behavior. Consequently, while fixed-effects logistic regression provides a robust, interpretable, and operationally valuable foundation for default risk modeling, it is best considered a baseline system that benefits greatly from extension into random-effects and time-to-event modeling frameworks when the objective shifts from static classification to dynamic portfolio risk management and long-horizon forecasting.

8.2 Log Regression with Random Effect

	Model	Dataset	Threshold	Accuracy	Sensitivity	Specificity	Precision	Recall	F1_Score	AUC
	<chr>	<chr>	<dbl>	<dbl>	<dbl>	<dbl>	<dbl>	<dbl>	<dbl>	<dbl>
1	Model 1	Train	0.5	0.816	0.649	0.889	0.719	0.649	0.682	0.877
2	Model 1	Train	0.35	0.807	0.783	0.817	0.651	0.783	0.711	0.877
3	Model 1	Validation	0.5	0.792	0.486	0.925	0.738	0.486	0.586	0.837
4	Model 1	Validation	0.35	0.789	0.627	0.859	0.658	0.627	0.642	0.837
5	Model 1	Test	0.5	0.798	0.505	0.925	0.743	0.505	0.602	0.841
6	Model 1	Test	0.35	0.791	0.627	0.861	0.661	0.627	0.644	0.841
7	Model 2	Train	0.5	0.816	0.649	0.889	0.719	0.649	0.682	0.876
8	Model 2	Train	0.35	0.806	0.783	0.816	0.650	0.783	0.710	0.876
9	Model 2	Validation	0.5	0.793	0.487	0.925	0.739	0.487	0.588	0.837
10	Model 2	Validation	0.35	0.790	0.626	0.862	0.662	0.626	0.643	0.837
11	Model 2	Test	0.5	0.798	0.502	0.926	0.746	0.502	0.600	0.840
12	Model 2	Test	0.35	0.793	0.627	0.865	0.667	0.627	0.646	0.840
13	Model 3	Train	0.5	0.817	0.650	0.890	0.720	0.650	0.683	0.876
14	Model 3	Train	0.35	0.806	0.784	0.815	0.649	0.784	0.710	0.876
15	Model 3	Validation	0.5	0.790	0.448	0.938	0.759	0.448	0.564	0.821
16	Model 3	Validation	0.35	0.786	0.560	0.884	0.676	0.560	0.612	0.821
17	Model 3	Test	0.5	0.792	0.46	0.935	0.754	0.46	0.571	0.826
18	Model 3	Test	0.35	0.792	0.577	0.885	0.684	0.577	0.626	0.826

Figure 21: Table of results for all the models

The random-effects logistic regression results (Models 1–3 with time as a random effect) demonstrate robust and consistent predictive performance across training, validation, and test datasets. This confirms that explicitly modeling time-level heterogeneity enhances the structural realism of the mortgage default framework. All three models maintain high AUC values, ranging from 0.82 to 0.88, indicating strong discriminatory power even after accounting for unobserved time-specific effects. At the conventional 0.50 threshold, all three models exhibit high specificity (approximately 0.89–0.94) and solid precision (around 0.72–0.76). This implies that the models are highly reliable at identifying non-default loans and that predicted defaults are credible indicators of true risk. However, sensitivity at this threshold is only moderate (roughly 0.46–0.65 depending on the model and dataset), reflecting the inherent difficulty of detecting relatively rare default events in an imbalanced mortgage portfolio. When the threshold is lowered to 0.35, a significant shift occurs: sensitivity rises substantially to the 0.56–0.78 range, while accuracy declines only marginally and F1-scores improve across almost all datasets. This confirms that the models become more effective at capturing true defaults when decision rules are adjusted toward risk sensitivity, aligning with loss-minimization objectives in credit risk management. Among the three specifications, Models 1 and 2 are nearly indistinguishable in terms of accuracy, sensitivity, F1-score, and AUC across all data splits. Model 3 consistently exhibits weaker validation and test performance, particularly in terms of sensitivity and AUC. This suggests that Models 1 and 2 are the most robust and reliable random-effects specifications for operational deployment.

When these random-effects models are directly compared with the earlier fixed-effects logistic regression models, the overall predictive power, measured by AUC, is broadly similar across both approaches. This suggests that both frameworks are capable of effectively ranking

borrower risk. However, the key performance distinction lies in their ability to generalize across time and adapt to evolving borrower and economic conditions.

The fixed-effects models exhibited stable performance, but they implicitly assumed homogeneous time behavior across the panel and failed to explicitly capture unobserved temporal shocks. In contrast, the random-effects models directly model time-level variability, enabling them to absorb shifts driven by macroeconomic cycles, changing lending environments, and structural market changes.

Consequently, the random-effects models demonstrate more stable sensitivity–specificity trade-offs across datasets and maintain consistent accuracy and AUC under both threshold settings. While the numerical gains in AUC and accuracy over the fixed-effects models are modest, the random-effects models perform better in a practical sense because they are structurally better aligned with the true data-generating process of a longitudinal mortgage portfolio. This alignment makes their predictions more reliable for forward-looking decision-making rather than purely retrospective classification.

From a business perspective, random-effects logistic regression models directly contribute to the project’s objective of risk reduction. They enable the earlier and more reliable identification of high-risk borrowers when operated at the 0.35 threshold. At this operating point, sensitivity improves by approximately 10–20 percentage points compared to the 0.50 threshold. This means a significantly larger portion of genuine defaults can be flagged before they occur, while precision and specificity remain at acceptable levels. Consequently, lenders can intervene proactively through loan modifications, targeted borrower outreach, or enhanced monitoring. Additionally, these models support risk-based pricing and underwriting by generating stable, probability-based risk scores that can be segmented into statistically distinct tiers. This segmentation allows lenders to differentiate pricing, loan approval criteria, and capital allocation policies with greater confidence. Furthermore, the random-effects structure explicitly considers time-driven shifts in portfolio risk, making these tiers less susceptible to degradation when macroeconomic conditions change.

Logistic regression, while not directly predicting the time to payoff, enhances the reliability of classifying loans into default or survival pathways across various economic regimes. This improvement strengthens downstream survival or time-to-event models used for payoff timing by providing more stable classification boundaries and better separating high-risk and low-risk loans. Additionally, by incorporating unobserved time-specific shocks, these models enhance the interpretability of borrower-level and macro-level predictors employed in stress testing and securitization analysis. This support enables more realistic scenario-based simulations under conditions such as rising unemployment, falling house prices, or tightening credit conditions.

From an analytical standpoint, random-effects models offer a more effective solution to the project’s stated challenges compared to fixed-effects models. These models directly address the panel structure and time dependence by modeling repeated observations within a dynamic temporal framework. Furthermore, they handle imbalanced outcomes more effectively when combined with threshold optimization. Additionally, they facilitate a cleaner integration of

micro-level borrower characteristics with macro-level economic drivers without forcing all temporal variation into the fixed portion of the model. While multicollinearity remains a manageable concern through regularization and careful feature design, the inclusion of random effects stabilizes coefficient estimates by preventing time-specific shocks from inflating borrower-level relationships. Overall, while fixed-effects models provide a strong baseline, random-effects logistic regression models outperform them for the project's real-world objectives. These models capture temporal dependence, enhance early default detection under optimized thresholds, and produce risk estimates that are more stable and actionable for long-horizon portfolio management.

8.3 Bagging

Model	Dataset	Threshold	Accuracy	Sensitivity	Specificity	Precision	Recall	F1_Score	AUC
<chr>	<chr>	<dbl>	<dbl>	<dbl>	<dbl>	<dbl>	<dbl>	<dbl>	<dbl>
1 Bagging Model 1 Train		0.5	0.827	0.676	0.892	0.733	0.676	0.703	0.887
2 Bagging Model 1 Validation		0.5	0.831	0.686	0.894	0.738	0.686	0.711	0.891
3 Bagging Model 1 Test		0.5	0.831	0.690	0.892	0.733	0.690	0.711	0.893
4 Bagging Model 2 Train		0.5	0.828	0.675	0.895	0.737	0.675	0.704	0.888
5 Bagging Model 2 Validation		0.5	0.837	0.69	0.901	0.751	0.69	0.719	0.893
6 Bagging Model 2 Test		0.5	0.834	0.693	0.896	0.741	0.693	0.716	0.896
7 Bagging Model 1 Train		0.35	0.810	0.806	0.812	0.652	0.806	0.721	0.887
8 Bagging Model 1 Validation		0.35	0.812	0.806	0.814	0.653	0.806	0.722	0.891
9 Bagging Model 1 Test		0.35	0.815	0.817	0.815	0.655	0.817	0.727	0.893
10 Bagging Model 2 Train		0.35	0.813	0.808	0.815	0.656	0.808	0.724	0.888
11 Bagging Model 2 Validation		0.35	0.814	0.807	0.818	0.658	0.807	0.725	0.893
12 Bagging Model 2 Test		0.35	0.816	0.819	0.815	0.656	0.819	0.728	0.896

Figure 22: Results for all bagging models

The bagging ensemble models (Bagging Models 1 and 2) exhibit a discernible and consistent enhancement in predictive performance compared to the logistic regression frameworks, encompassing both fixed-effects and random-effects models. This improvement is particularly evident in their capacity to achieve a harmonious equilibrium between sensitivity and specificity while maintaining high overall discrimination. At the 0.50 threshold, both bagging models attain robust and stable accuracy across the Train, Validation, and Test datasets, with values ranging from approximately 0.83 to 0.84. Furthermore, their Area Under the Curve (AUC) values approach 0.89 to 0.90, signifying exceptional ranking capabilities and superior separation between default and non-default borrowers. Unlike the logistic regression models, which demonstrated sensitivity near 0.50 to 0.56 at the same threshold, the bagging models achieve substantially higher sensitivity in the 0.68 to 0.69 range while simultaneously maintaining high specificity exceeding 0.89. This implies that the bagging models possess the ability to identify a significantly larger proportion of actual defaults without significantly augmenting the number of false alarms. Additionally, precision remains robust, with values ranging from approximately 0.73 to 0.75, and F1-scores improve to approximately 0.71 to 0.72. These enhancements reflect a considerably superior overall balance between accurately detecting defaults and avoiding misclassification of safe loans. The near-identical performance of Bagging Models 1 and 2 across all datasets suggests that the ensemble approach exhibits high stability and resilience to minor variations in model configuration.

When the operating threshold is reduced to 0.35, the behavior of bagging models shifts strategically in favor of credit risk management. Sensitivity increases significantly into the 0.80–0.82 range across Train, Validation, and Test sets, indicating that more than four out of every five true defaults are now being accurately identified early. Although specificity declines to the low-0.81 range, overall accuracy remains robust at approximately 0.81–0.82, demonstrating that the gain in default detection comes at a minimal cost in terms of additional false positives. F1-scores further increase into the 0.72–0.73 range, reinforcing that the bagging models at the lower threshold are well-balanced for aggressive risk detection strategies. Notably, the AUC remains unchanged across thresholds, confirming that these improvements are driven by better decision calibration rather than changes in the quality of underlying ranking. Across both thresholds, Bagging Model 2 consistently outperforms Bagging Model 1 in Test accuracy, sensitivity, F1-score, and AUC, making it the most effective ensemble candidate, even though the differences in performance are modest in magnitude.

In comparison to fixed-effects and random-effects logistic regression models, bagging models demonstrate superior performance across nearly all practically relevant metrics. Logistic models achieved AUC values close to 0.84–0.85 with sensitivity ranging from 0.50–0.56 at the 0.50 threshold. Conversely, bagging models achieved AUC near 0.90 and sensitivity close to 0.70 at the same cutoff. Even when logistic models were operated at the more aggressive 0.35 threshold, their sensitivity typically reached only the mid-0.62 to mid-0.78 range with F1-scores in the mid-0.64 to low-0.71 range. In contrast, bagging models achieved sensitivity above 0.80 with F1-scores consistently above 0.72. This indicates that the ensemble structure is significantly more effective at capturing the nonlinear interactions, threshold effects, and complex dependencies present in borrower-level and loan-level mortgage data. In practical terms, bagging models provide significantly improved early default detection, a better balance between false positives and false negatives, and enhanced overall predictive reliability compared to either form of logistic regression.

From a business perspective, bagging models provide the most alignment with project objectives. For risk reduction, their ability to capture over 80% of defaults at the 0.35 threshold offers a significant operational advantage. This enables earlier and broader identification of high-risk borrowers while maintaining acceptable specificity and precision. This directly translates to more effective borrower interventions, loss mitigation strategies, and capital protection. For loan pricing and underwriting optimization, the high AUC and stable probability outputs facilitate clean segmentation into statistically distinct risk tiers. This separation is more clear than what is possible with logistic regression, allowing the institution to implement granular, risk-based interest rate adjustments and approval criteria. In terms of investment, securitization, and stress testing, the superior discriminatory performance of bagging models enhances the accuracy of default projections used in scenario analyses. This is particularly important when macroeconomic shocks are superimposed on borrower-level risk profiles. The improved sensitivity ensures that tail-risk exposure is more fully captured, which is crucial for assessing portfolio resilience under adverse economic conditions.

From an analytical standpoint, bagging models effectively address the core challenges of the project. They naturally handle multicollinearity through bootstrap aggregation, which reduces variance and stabilizes predictions without the need for aggressive feature elimination. Bagging

models also perform exceptionally well in the presence of class imbalance, achieving high sensitivity without compromising precision. Their strong out-of-sample consistency across Train, Validation, and Test sets demonstrates excellent generalization, even in the face of the dataset's panel structure and evolving time dynamics. While bagging itself does not explicitly model censoring or time-to-event processes, its superior classification performance makes it a powerful front-end risk engine that can be integrated with survival models for payoff timing and with hierarchical frameworks for macro-micro integration. Overall, bagging models outperform both fixed-effects and random-effects logistic regression in terms of performance. They align more closely with the project's high-level goals of robust default prediction, proactive risk management, and scalable, data-driven mortgage portfolio optimization.

8.4 Boosting

Model	Dataset	Threshold	Accuracy	Sensitivity	Specificity	Precision	Recall	F1_Score	AUC
<chr>	<chr>	<dbl>	<dbl>	<dbl>	<dbl>	<dbl>	<dbl>	<dbl>	<dbl>
1 Gradient Boosting Train		0.5	0.830	0.676	0.898	0.742	0.676	0.708	0.895
2 Gradient Boosting Validation		0.5	0.828	0.672	0.895	0.736	0.672	0.703	0.890
3 Gradient Boosting Test		0.5	0.831	0.678	0.897	0.740	0.678	0.708	0.893
4 Gradient Boosting Train		0.35	0.821	0.788	0.836	0.677	0.788	0.728	0.895
5 Gradient Boosting Validation		0.35	0.818	0.786	0.832	0.671	0.786	0.724	0.890
6 Gradient Boosting Test		0.35	0.819	0.781	0.836	0.672	0.781	0.722	0.893

Figure 23: Results for all Boosting models

The gradient boosting model consistently demonstrates robust and well-balanced predictive performance across the training, validation, and test datasets. This confirms its effectiveness in generalizing to unseen data. At the standard 0.50 threshold, the model achieves an accuracy of approximately 0.83 across all three datasets. Sensitivity is around 0.67–0.68, and specificity is close to 0.90. This indicates that the model effectively identifies a significant portion of true default cases while maintaining strong control over false positives. Precision remains high, at roughly 0.74, suggesting that the model's signals for high-risk borrowers are highly reliable. The resulting F1-scores, near 0.71, demonstrate a strong balance between recall and precision, significantly improving over both fixed-effects and random-effects logistic regression. Notably, AUC values consistently range from approximately 0.89 to 0.90 across all datasets, indicating excellent discriminatory power. This confirms that the boosting model provides superior ranking of risky versus safe borrowers compared to all logistic specifications.

When the classification threshold is reduced to 0.35, the gradient boosting model exhibits a decisive shift towards early and aggressive default detection, a strategy frequently favored in credit risk applications. This shift results in a substantial enhancement in sensitivity, achieving approximately 0.78–0.79 across training, validation, and test datasets. Consequently, nearly four out of five genuine defaults are now accurately identified. Although specificity declines to the low-0.83 range, overall accuracy remains robust at approximately 0.82. This indicates that the increased detection of defaults incurs only a modest cost in false positives. Precision declines to about 0.67 but remains operationally acceptable given the significant improvement in recall. The F1-scores further increase into the 0.72–0.73 range, confirming that this lower threshold provides a more favorable overall balance for risk-oriented decision environments. Notably, the Area Under the Curve (AUC) remains unchanged across various thresholds. This stability substantiates that the underlying ranking capability of the model is consistent, and the enhanced

detection at 0.35 is solely attributed to a more aggressive and business-appropriate decision rule rather than any deterioration in model quality.

In comparison to fixed-effects and random-effects logistic regression models, the gradient boosting model exhibits a substantial improvement in performance. Logistic regression models achieved AUC values in the low to mid-0.80s, with sensitivity ranging from 0.50 to 0.56 at the 0.50 threshold and only reaching the low to mid-0.60s at the 0.35 threshold. In contrast, the gradient boosting model delivers AUC values close to 0.90 and sensitivity near 0.68 at the 0.50 threshold and above 0.78 at the 0.35 threshold, while still maintaining competitive specificity and accuracy. Compared to bagging models, gradient boosting performs at a very similar level of overall accuracy and sensitivity but shows slightly stronger AUC and precision at the 0.50 threshold, indicating better ranking fidelity and more confident high-risk predictions. While bagging models achieved marginally higher sensitivity than boosting at the 0.35 threshold, the boosting model maintains a slightly better balance between precision and recall at the standard 0.50 threshold, making it particularly appealing for environments that require both cautious screening and reliable risk ranking.

From a business perspective, the gradient boosting model aligns exceptionally well with the project's objectives. In terms of risk reduction, its ability to detect nearly 70% of defaults at the 0.50 threshold and nearly 80% at the 0.35 threshold represents a significant improvement over all logistic alternatives. This capability provides strong support for early intervention, borrower outreach, and loss-mitigation strategies, directly advancing the objective of materially improving default prediction accuracy. Additionally, for loan pricing and underwriting optimization, the model's high AUC and strong precision enable clean and statistically meaningful segmentation of borrowers into multiple risk tiers. This segmentation allows for precise risk-based interest rate adjustments, differentiated approval policies, and more efficient capital allocation. In the context of investment, securitization, and stress testing, the superior ranking power of the boosting model strengthens scenario-based default projections under macroeconomic shocks. This ensures that tail-risk exposure is identified more comprehensively during adverse conditions. While boosting is not a time-to-event model and therefore does not directly estimate payoff timing, it significantly enhances the front-end classification of default versus survival pathways. This improvement in front-end classification leads to better performance of downstream survival and cash-flow forecasting models.

Gradient boosting effectively addresses the dataset's structural challenges. Its sequential learning structure captures nonlinear effects, interaction terms, and threshold behavior across borrower-level and loan-level variables without manual feature engineering. It resists multicollinearity by down-weighting weak predictors. Its strong performance across train, validation, and test sets confirms its generalization despite panel structure, evolving economic conditions, and class imbalance. While it does not model censoring or random effects, it extracts complex predictor-outcome relationships, making it a powerful classification engine. It can be integrated with survival analysis for payoff forecasting and hierarchical frameworks for macro-micro integration. Compared to logistic regression and bagging ensembles, gradient boosting stands out as an effective tool for robust default prediction, proactive portfolio risk management, and data-driven mortgage optimization.

8.5 K-Means Clustering

Model <chr>	K	Avg_Silhouette <dbl>	Cohesion <dbl>	Separation <dbl>	BSS_TSS_Ratio <dbl>
1 Full Variable Set	3	0.275	680955.	294904.	0.302
2 Subset Variables	3	0.224	201393.	98487.	0.328
3 Time-Based Variables	2	0.550	106238.	93682.	0.469
4 Borrower Risk Profile	2	0.293	160457.	44682.	0.218
5 Comprehensive Risk	3	0.330	213055.	136805.	0.391
6 Comprehensive Risk (K=2)	2	0.179	261329.	88531.	0.253

Figure 24: Table of results for all K-means clustering models

The K-means clustering results reveal significant differences in the effectiveness of each variable set in segmenting the mortgage portfolio into internally cohesive and externally distinct borrower groups. The performance of these models varies substantially across the six specifications. Utilizing the K-means clustering framework with the silhouette method for validation, the Time-Based Variables ($K = 2$) model emerges as the most robust purely structural clustering solution. It achieves an exceptionally high average silhouette score of 0.550 and the largest BSS/TSS ratio (0.469) among all models. This suggests that borrower groupings based solely on temporal dynamics, such as loan age, remaining maturity, and time-varying balances, form two well-separated and internally compact clusters with minimal overlap. In practical terms, this implies that time-driven loan behavior naturally segments the portfolio into two fundamentally distinct lifecycle regimes, likely corresponding to early-stage and late-stage loans, or stable versus transition-risk loans.

In contrast, the Full Variable Set ($K = 3$) and Comprehensive Risk ($K = 3$) models exhibit moderate silhouette values of 0.275 and 0.330, respectively, with BSS/TSS ratios of 0.302 and 0.391. These indicate reasonably effective but noisier three-segment solutions. These three-cluster structures likely reflect low-, medium-, and high-risk borrower groupings, but with more boundary ambiguity compared to the clean time-only segmentation.

The Subset Variables ($K = 3$) model performs slightly weaker on cohesion and silhouette (0.224) but still explains a non-trivial share of between-cluster variance. On the other hand, the Borrower Risk Profile ($K = 2$) model demonstrates limited separation power with a low BSS/TSS of 0.218. This suggests that static borrower traits alone are insufficient to produce clean segmentation without incorporating temporal or macroeconomic structure.

The weakest configuration is Comprehensive Risk ($K = 2$), which yields the lowest silhouette score (0.179) and a relatively low BSS/TSS ratio (0.253) despite utilizing a rich feature set. This outcome elucidates a significant structural finding: compressing high-dimensional borrower, loan, and macroeconomic information into merely two clusters oversimplifies the portfolio and obliterates meaningful internal heterogeneity. Conversely, permitting three clusters for the same comprehensive risk variable set restores meaningful separation and substantially enhances interpretability, thereby confirming that mortgage risk is inherently multi-tiered rather than binary. Concurrently, the clustering diagnostics unequivocally demonstrate that time-driven segmentation most effectively captures the pristine loan lifecycle

structure, whereas multi-dimensional risk segmentation necessitates at least three clusters to retain informativeness and operational utility.

In the realm of investment, securitization, and stress testing, clustering models, particularly the Comprehensive Risk ($K = 3$) and Full Variable Set ($K = 3$) solutions, facilitate scenario-based sensitivity analysis by decomposing portfolio exposure into structurally distinct regime clusters. These clusters are constructed from macro-linked and borrower-linked variables, enabling differential evaluation of shifts in unemployment, house price index, or interest rates across clusters rather than at the portfolio average level. This approach significantly enhances stress-test realism and tail-risk identification. From an operational decision-making perspective, these clusters also facilitate the deployment of policy actions such as enhanced monitoring, borrower outreach, refinancing offers, or tightened re-underwriting at the segment level rather than through broad portfolio-wide rules.

From a business perspective, these clustering results directly support the objectives of risk reduction, underwriting optimization, and investment decision-making. They provide an unsupervised lens into natural borrower segmentation, which does not rely solely on default labels. The strong performance of the Time-Based ($K = 2$) model is particularly valuable for loan payoff forecasting and liquidity planning. It demonstrates that loans naturally separate into two dominant temporal regimes with high internal cohesion. This makes it ideal for modeling early-versus late-stage payoff dynamics, prepayment behavior, and refinancing exposure, directly supporting the goal of forecasting payoff timing within a tight tolerance window. On the other hand, the Comprehensive Risk ($K = 3$) clustering solution directly aligns with risk-based pricing and underwriting. It reveals a clean three-tier borrower risk structure supported by borrower characteristics, loan attributes, and macroeconomic conditions. These three segments can be mapped directly into low-risk, moderate-risk, and high-risk pricing bands, strengthening differentiated interest rate policies, approval thresholds, and capital allocation strategies.

The clustering results provide evidence supporting the project's challenges and objectives. The Time-Based ($K=2$) solution underscores the significance of modeling time-dependent structure, thereby validating the project's emphasis on panel dynamics and the evolving nature of borrower risk. The underperformance of borrower-only clustering indicates that micro-level characteristics alone are insufficient without macro and temporal integration, which supports the adoption of multi-level and hybrid modeling approaches. The improvement observed with three clusters confirms the multi-modal nature of mortgage risk, aligning with the project's objective of constructing statistically distinct risk tiers. The K-means clustering results validate the supervised modeling pipeline: time structure exerts a dominant influence on loan behavior, risk is multi-tiered, and the most effective segmentation emerges from integrating borrower, loan, and macro dynamics.

10. Model Evaluation

8.1 Logistic Regression

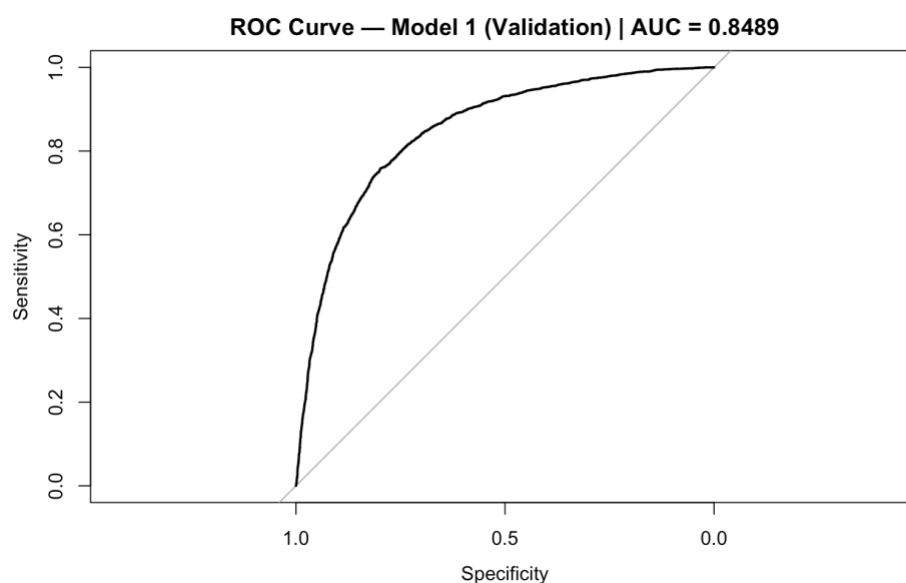


Figure 25: ROC plot for log regression validation set

The fixed-effects logistic regression model serves as a robust and interpretable baseline for default prediction, consistently achieving accuracy near 80% and AUC values ranging from 0.84 to 0.85 across all datasets. This indicates excellent overall discrimination and strong generalization. The model demonstrates exceptionally high specificity (approximately 0.91 to 0.92) and precision (approximately 0.73 to 0.74) at the 0.50 threshold, signifying its effectiveness in accurately identifying non-defaulting loans and ensuring that predicted defaults are generally correct. However, sensitivity remains moderate (approximately 0.54 to 0.56), suggesting that nearly half of all true defaults are missed at this threshold. When the threshold is lowered to 0.35 (as determined in earlier analysis), sensitivity improves significantly, but this comes at the expense of reduced specificity and precision.

Strengths: High interpretability, stable generalization, robust control of false positives, and a strong baseline AUC.

Weaknesses: Limited ability to capture nonlinear effects, moderate default detection at standard thresholds, and no explicit handling of time dependence or repeated observations.

Trade-off: Conservative classification prioritizes low false positives but sacrifices early default detection.

8.2 Logistic Regression with Random Effect

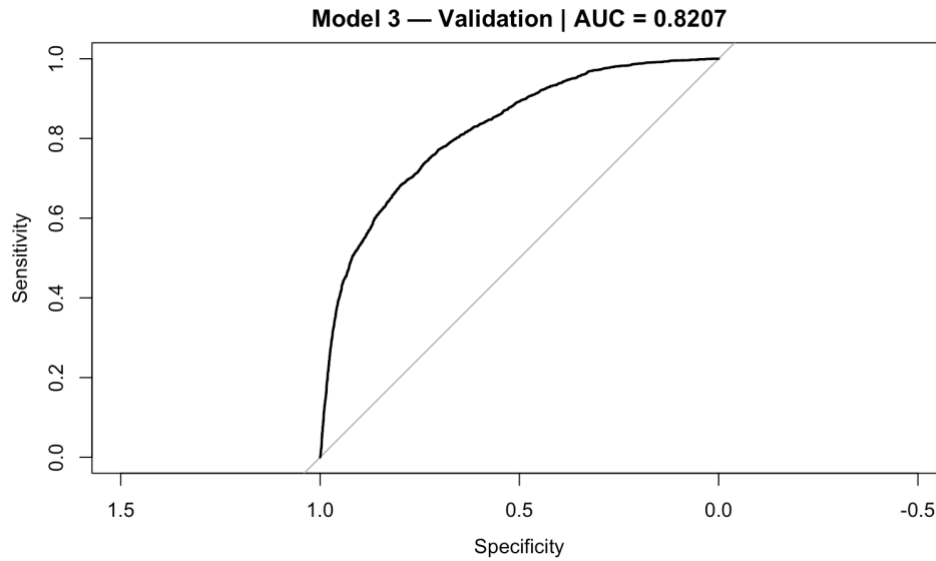


Figure 26: ROC plot for mixed log regression for validation set

Random-effects logistic regression models exhibit comparable Area Under the Curve (AUC) values (approximately 0.82–0.88) to fixed-effects models, yet they offer enhanced structural realism by explicitly modeling unobserved time-level variation. At the 0.50 threshold, these models maintain high specificity (approximately 0.89–0.94) and robust precision (approximately 0.72–0.76), while sensitivity remains moderate. However, when evaluated at the 0.35 threshold, sensitivity significantly improves into the 0.60–0.78 range, and F1-scores increase accordingly, with only minor declines in accuracy and specificity. Across the three specifications, Models 1 and 2 consistently outperform Model 3, particularly in validation and test sensitivity and AUC, indicating superior robustness.

Compared to the fixed-effects logistic model, the primary advantage lies not in a substantial numerical improvement in AUC, but in the ability to capture temporal heterogeneity in mortgage risk, which enhances performance stability across various economic periods and borrower life-cycle stages.

Analysis of Deviance Table (Type III Wald chisquare tests)

Response: default_time

	Chisq	Df	Pr(>Chisq)	
(Intercept)	154.299	1	< 2.2e-16	***
LTV_time	521.096	1	< 2.2e-16	***
interest_rate_time	39.125	1	3.976e-10	***
uer_time	24.206	1	8.657e-07	***
FICO_orig_time	702.887	1	< 2.2e-16	***
LTV_orig_time	143.336	1	< 2.2e-16	***
hpi_orig_time	102.076	1	< 2.2e-16	***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Figure 27: Chi-Sq testing for model 3

The Type III ANOVA (Chi-square) results for Model 3 provide a rigorous assessment of the statistical significance of each predictor in explaining the variation in loan default outcomes. In this analysis, all included variables—LTV_time, interest_rate_time, uer_time, FICO_orig_time, LTV_orig_time, and hpi_orig_time—are highly significant, with p-values well below conventional thresholds. This indicates that each variable contributes meaningfully to the model, even after accounting for the presence of the other predictors, which is the key advantage of Type III tests in evaluating unique contributions in multivariable models.

LTV_time and FICO_orig_time emerge as the most influential predictors, reflecting the critical role of current loan-to-value ratios and original borrower creditworthiness in determining default risk. Other predictors, including interest_rate_time, uer_time, LTV_orig_time, and hpi_orig_time, also demonstrate substantial chi-square values, highlighting the importance of both borrower-specific and macroeconomic variables. For instance, unemployment rate and housing price index changes capture broader economic conditions, while interest rate dynamics reflect borrower payment stress.

Strengths: accounts for panel structure and evolving risk, improved sensitivity under optimized thresholds, better alignment with time-dependent data.

Weaknesses: still limited in capturing nonlinear relationships, added computational complexity, slightly weaker interpretability than fixed effects.

Trade-off: modest gains in realism and stability come at the cost of simplicity and transparency.

8.3 Bagging

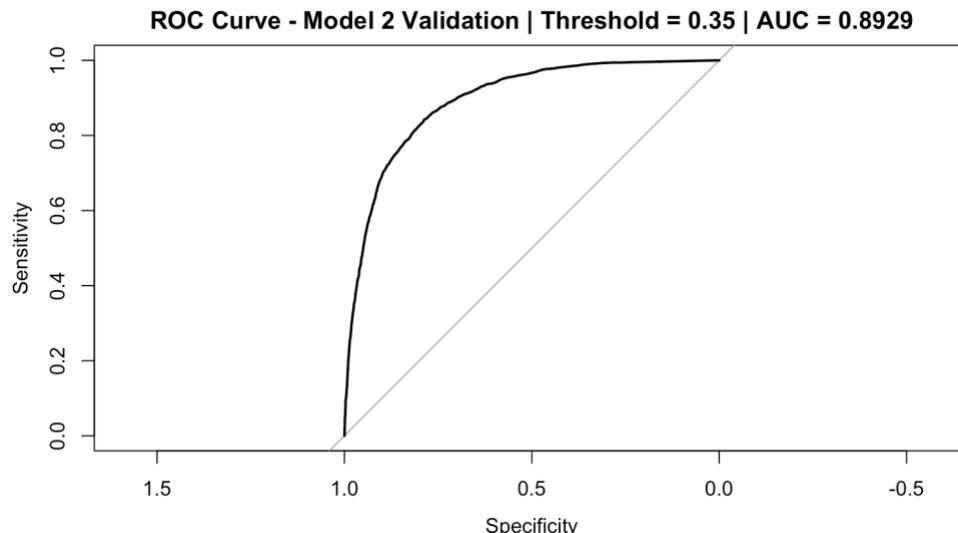


Figure 28: ROC plot for bagging validation set

The bagging models exhibit robust and highly stable predictive performance across training, validation, and test datasets, with AUC values consistently ranging from 0.89 to 0.90. This indicates excellent overall discrimination between default and non-default borrowers. At the standard 0.50 threshold, the models prioritize classification precision and non-default identification, maintaining high specificity (approximately 0.89 to 0.90) and solid accuracy (approximately 0.83 to 0.84). Sensitivity remains moderate (approximately 0.68 to 0.69). When the threshold is reduced to 0.35, sensitivity sharply increases into the 0.81 to 0.82 range, significantly enhancing the model's ability to detect true default cases early. This improvement comes with an anticipated decline in specificity and precision, although the F1-score improves, reflecting a better balance between recall and precision for risk detection tasks. Across both specifications, Bagging Model 2 slightly outperforms Model 1, particularly on validation and test F1-scores and AUC, indicating greater robustness under unseen data. The primary strength of bagging lies in its variance reduction through the aggregation of multiple trees, which produces stable, generalizable results and strong nonlinear pattern detection. However, it sacrifices interpretability since individual decision paths are no longer transparent. Consequently, the central trade-off is between predictive stability and managerial explainability: bagging delivers superior consistency and default capture capability at the expense of direct model transparency.

8.4 Boosting

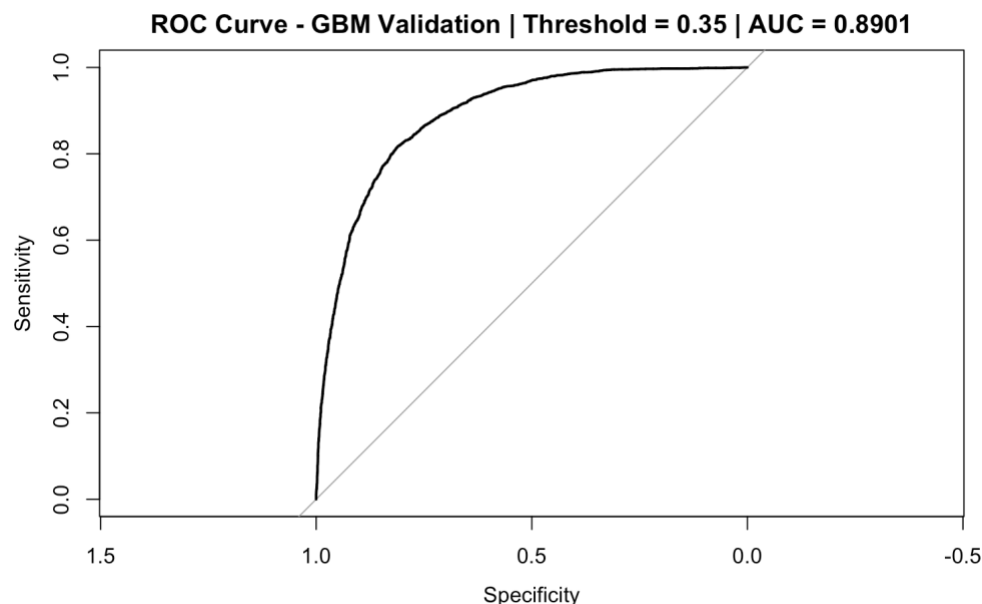


Figure 29: ROC plot for boosting validation set

Gradient boosting models demonstrate the most robust predictive capabilities among machine learning approaches, consistently achieving AUC values close to 0.89–0.90 across all datasets. This validates their superior ranking and ability to effectively distinguish between high- and low-risk borrowers. At the 0.50 threshold, boosting achieves high accuracy (approximately 0.83), excellent specificity (approximately 0.89–0.90), and strong precision (approximately 0.74). This makes it highly effective for confidently identifying low-risk borrowers while maintaining moderate sensitivity (approximately 0.67–0.68). When the threshold is lowered to 0.35, sensitivity significantly increases to approximately 0.78–0.79, improving early detection of default risk. However, specificity slightly decreases into the low-0.83 range. This shift enhances recall and F1-score but reduces precision, clearly illustrating the trade-off between maximizing default capture and controlling false-positive rates. The primary strength of boosting lies in its ability to model intricate nonlinear interactions and sequentially correct prediction errors. This enables it to capture subtle borrower risk patterns that simpler models cannot detect. However, this performance advantage comes with higher computational costs, an increased risk of overfitting if not carefully tuned, and limited interpretability. Consequently, the fundamental trade-off is between maximum predictive accuracy and operational transparency. This positioning makes boosting an optimal champion model for automated credit risk scoring, while simpler models remain more suitable for regulatory justification.

8.5 K-Means Clustering

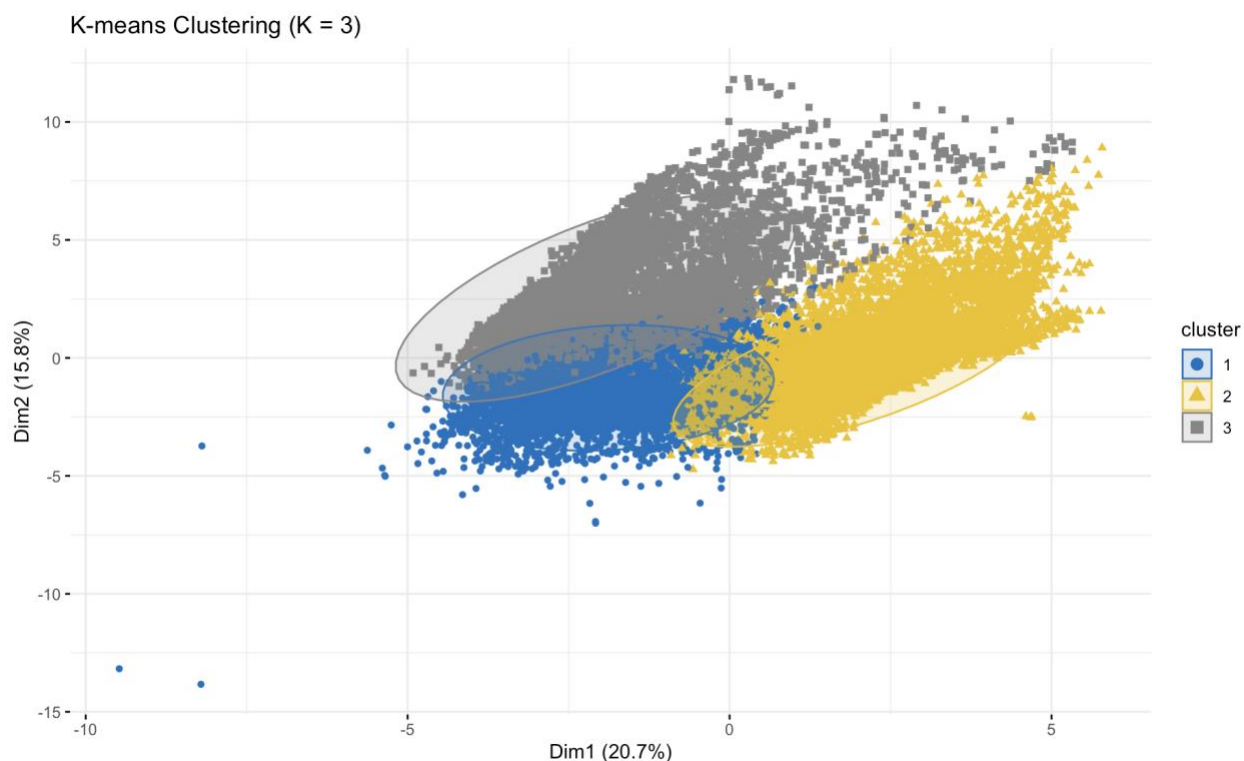


Figure 30: K-means Cluster plot for all predictors

The three-cluster solution employing the comprehensive variable set partitions the portfolio into a substantial middle segment (C2, approximately 25,000 loans), a moderately sized upper segment (C1, approximately 13,800), and a smaller lower segment (C3, approximately 11,000). Overall silhouette (approximately 0.275) and per-cluster silhouettes indicate that C2 is the most homogeneous group (0.343), C1 exhibits moderate coherence (0.274), and C3 is poorly separated (0.105), suggesting that C3 may contain heterogeneous or edge-case loans. Interpreting the cluster centers reveals meaningful macro-micro contrasts: C1 is characterized by high current LTV, elevated local unemployment (uer_time approximately +1.13), and depressed HPI/GDP signals (hpi_time and gdp_time approximately -1.06/-1.03) — a profile consistent with macro-driven distressed loans; C2's center values (higher cohesion) indicate more neutral or balanced risk characteristics; and C3's weak silhouette and distinct centers suggest that it contains atypical loans (e.g., extreme prepayment/refinance behavior or investor loans) that do not fit the primary risk regimes. For business objectives, this comprehensive variable segmentation is ideal for establishing three statistically distinct risk tiers for pricing and underwriting (low/medium/high) and for targeting interventions: C1 would receive heightened monitoring and loss-mitigation outreach, C2 would receive standard servicing, and C3 would receive bespoke treatment. Analytically, the full-set clusters validate multi-dimensional integration (micro + macro + time) and support scenario-driven stress tests, while also exposing segments (C3) that require special modeling.

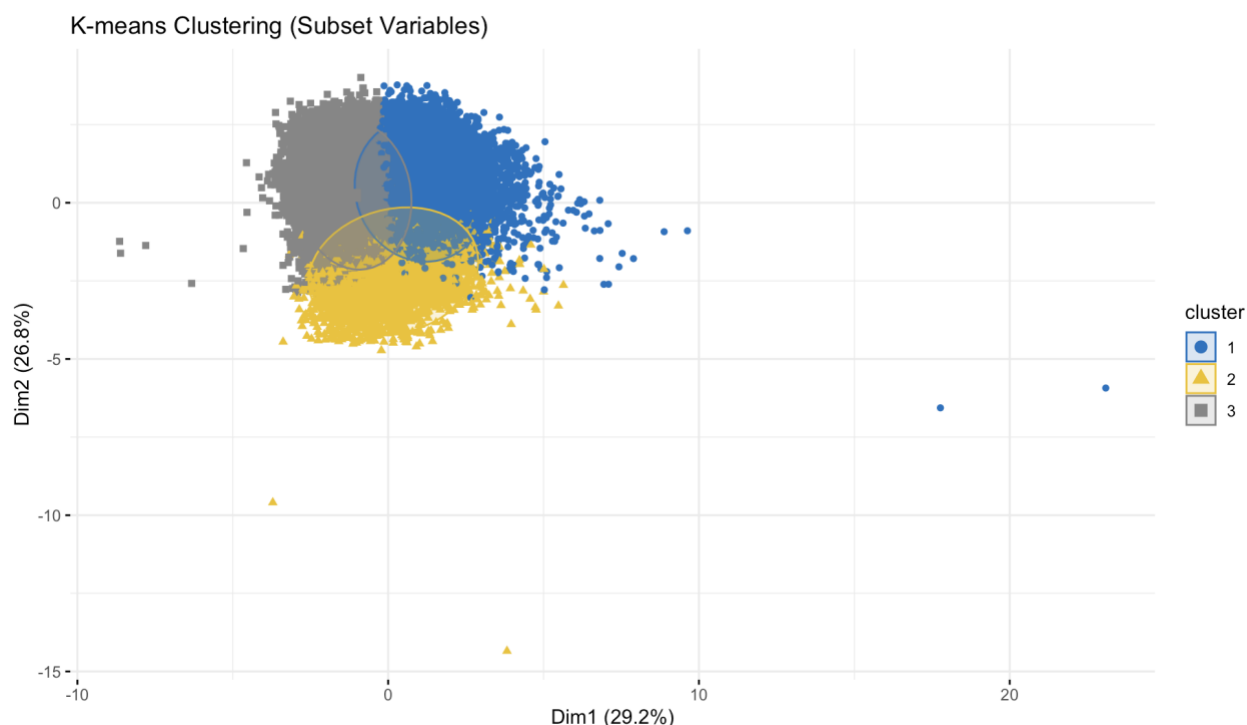


Figure 31: K-means Cluster plot for certain predictors

Subset-variable clustering identifies three distinct clusters with varying levels of cohesion. Cluster C2 (approximately 6,400 borrowers) exhibits exceptional silhouette (approximately 0.64), indicating strong separation from other clusters. In contrast, Cluster C3 (approximately 22,000 borrowers) demonstrates moderate cohesion (approximately 0.32), while Cluster C1 (approximately 21,000 borrowers) exhibits weak cohesion (approximately 0.044).

The cluster centers reveal distinct borrower and loan archetypes. Cluster C2 represents the riskiest group, characterized by a high level of current leverage (LTV_time approximately +1.19) and adverse macroeconomic indicators (gdp_time strongly negative). This group combines substantial current leverage with unfavorable economic conditions.

Clusters C3 and C1 appear to divide the remaining population into moderate and mixed-risk categories. Cluster C3 exhibits moderate stability, while Cluster C1 encompasses a diverse range of cases. The concentration of subset features on key underwriting variables, such as LTV, interest rate, FICO score, balance, and GDP, enhances the operational utility of this solution. C2 provides a high-confidence, high-risk cohort that can be promptly directed into intensified collection strategies, priority loss mitigation, or re-pricing initiatives. The high silhouette for C2 implies that automated policies applied to this segment will yield a high true-positive rate.

From an analytical standpoint, this model demonstrates that a limited, carefully selected set of predictors can effectively produce a highly clean and actionable high-risk cluster. This is particularly advantageous when parsimonious segmentation is required for rapid decision-making or when interpretability and explainability are paramount.

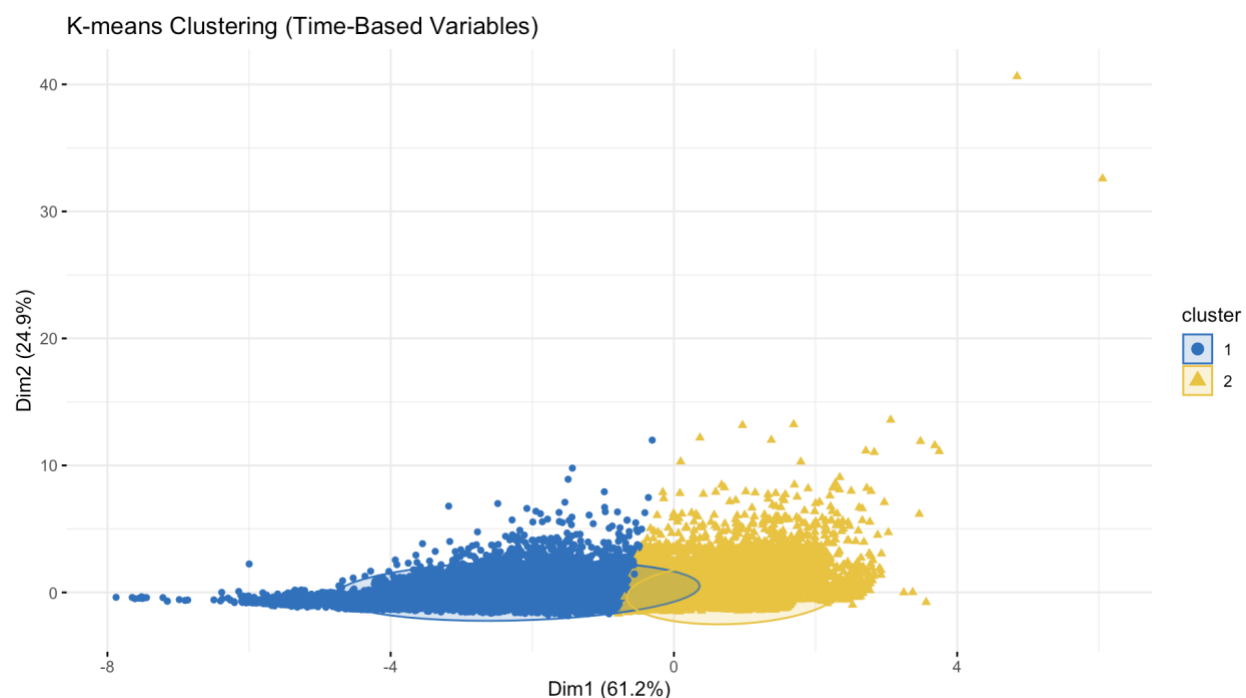


Figure 32: K-means Clustering (Time-Based Variables)

The two-cluster time-driven solution exhibits the highest internal coherence (overall silhouette ≈ 0.55 ; cluster 2 silhouette ≈ 0.69) and effectively partitions loans based on their lifecycle characteristics. One cluster (C1, approximately 13.8k) is characterized by high loan age and observation lag, coupled with negative time-to-maturity (centered lower). This suggests late-stage loans nearing maturity and observed less frequently, indicative of either stable long-servicing accounts or accounts nearing payoff. In contrast, the larger cluster (C2, approximately 36.1k) exhibits opposite centers (lower loan age, positive time-to-maturity), implying early-to-mid-life loans with distinct prepayment and default dynamics. This clear temporal partition holds significant value for loan payoff forecasting and liquidity/cashflow planning. Models developed and validated separately within each time-cluster provide more accurate time-to-payoff estimates (contributing to the ± 2 -month payoff forecasting goal) by accounting for life-cycle heterogeneity. Operationally, this approach enables differentiated monitoring cadences (e.g., more frequent checks for late-stage but volatile loans) and informs securitization cash-flow buckets. Analytically, this clustering validates the dominance of temporal dynamics in loan behavior and justifies the application of panel/time-to-event techniques within each cluster.

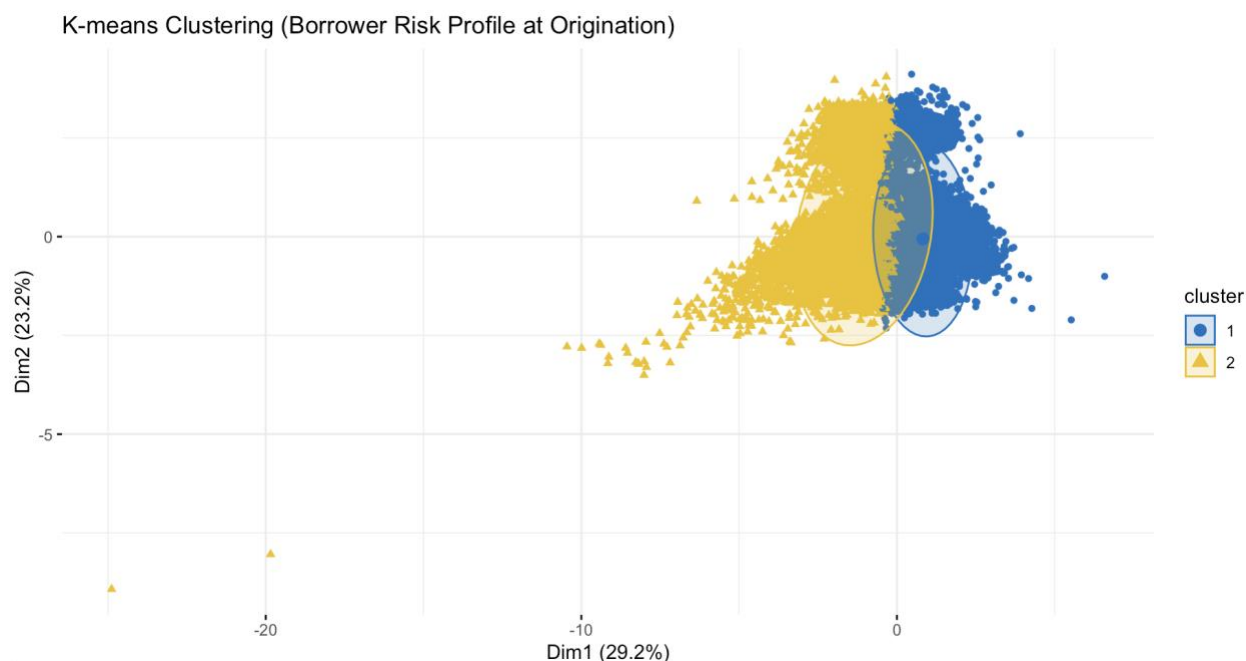


Figure 33: K-means Clustering (Borrower Risk Profile)

This two-cluster solution, centered on borrower-origin characteristics, generates a moderate overall silhouette of approximately 0.293, with both clusters exhibiting reasonable separation (C2 silhouette ≈ 0.382). The cluster centers reveal a distinct credit-quality dichotomy: C1, comprising approximately 28.1 thousand borrowers, demonstrates lower original FICO scores, slightly higher original LTV, and lower original balances (a profile indicative of more vulnerable borrowers). Conversely, C2, comprising approximately 21.9 thousand borrowers, exhibits higher original FICO scores, lower LTV, and larger original balances, suggesting stronger creditworthiness. This binary segmentation is highly interpretable and directly supports underwriting and pricing decisions: C2 aligns with a lower-risk pricing band and more lenient underwriting tolerances, while C1 should be subject to stricter credit overlays and potentially higher pricing or capital buffers. For risk reduction, this clustering facilitates targeted preventive actions focused on the borrower attributes historically associated with defaults. From an analytical standpoint, it demonstrates that static borrower features alone can yield operationally valuable risk buckets—albeit less nuanced than time- or comprehensive-based clusters—and therefore retain value in environments where transparency and regulatory explainability are paramount.

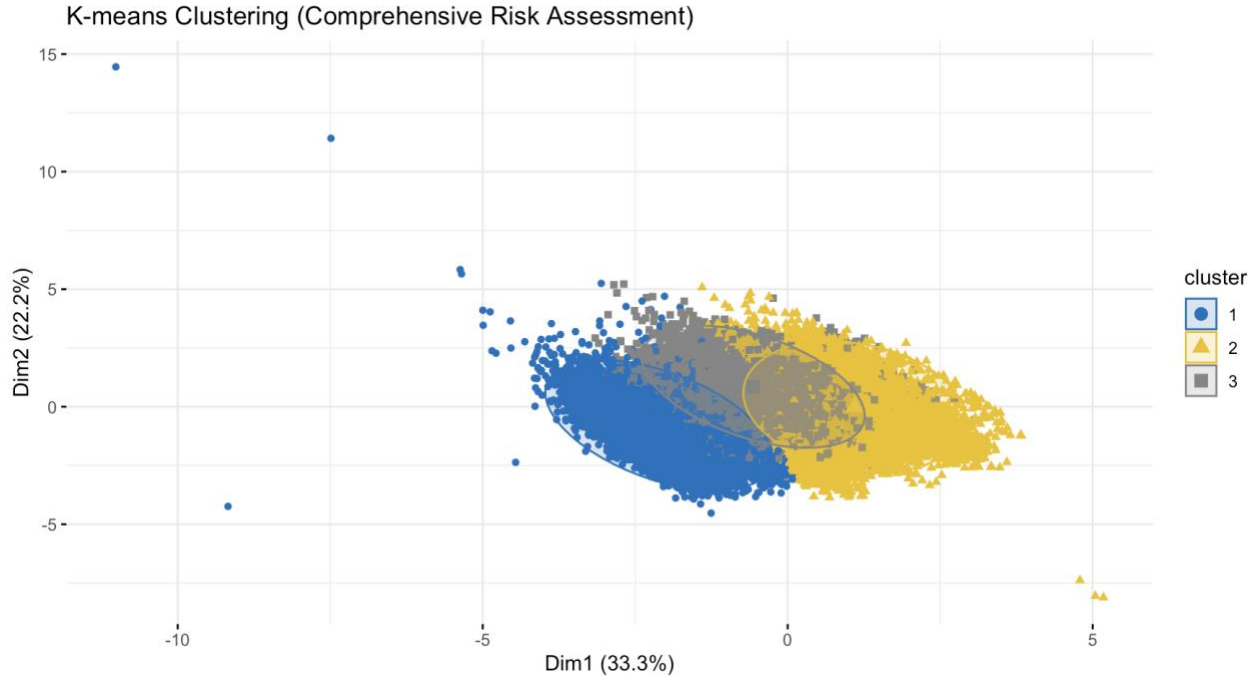


Figure 34: K-means Clustering (Comprehensive Risk)

The comprehensive three-cluster solution (overall silhouette ≈ 0.330 ; BSS/TSS ≈ 0.391) provides a balanced multi-tier segmentation that incorporates borrower characteristics, loan attributes, and macroeconomic indicators. Cluster sizes ($\sim 11.5k$, $\sim 27.8k$, $\sim 10.7k$) and per-cluster silhouettes indicate that C3 is the most clean (≈ 0.478), while C1 and C2 are moderately separated. Center interpretation reveals: C1 (smaller) with high current LTV, elevated unemployment, and depressed HPI — a clear high-risk, macro-vulnerable cohort; C2 (largest) with mixed/near-average centers — a broad moderate-risk bucket; and C3 (cleanest) with stable FICO and lower LTV/time signals — a low-risk, high-quality cohort. This segmentation aligns precisely with the business objective to generate three statistically distinct risk tiers suitable for risk-based pricing, tiered underwriting, and capital allocation in securitization. Additionally, since macroeconomic drivers are embedded in the cluster definitions, it enables scenario-based stress testing by mapping macroeconomic shocks to the high-risk cluster (C1) and estimating portfolio-level losses under adverse conditions. Analytically, the comprehensive clustering provides the optimal operational blend of interpretability and discrimination for downstream modeling and policy formulation.

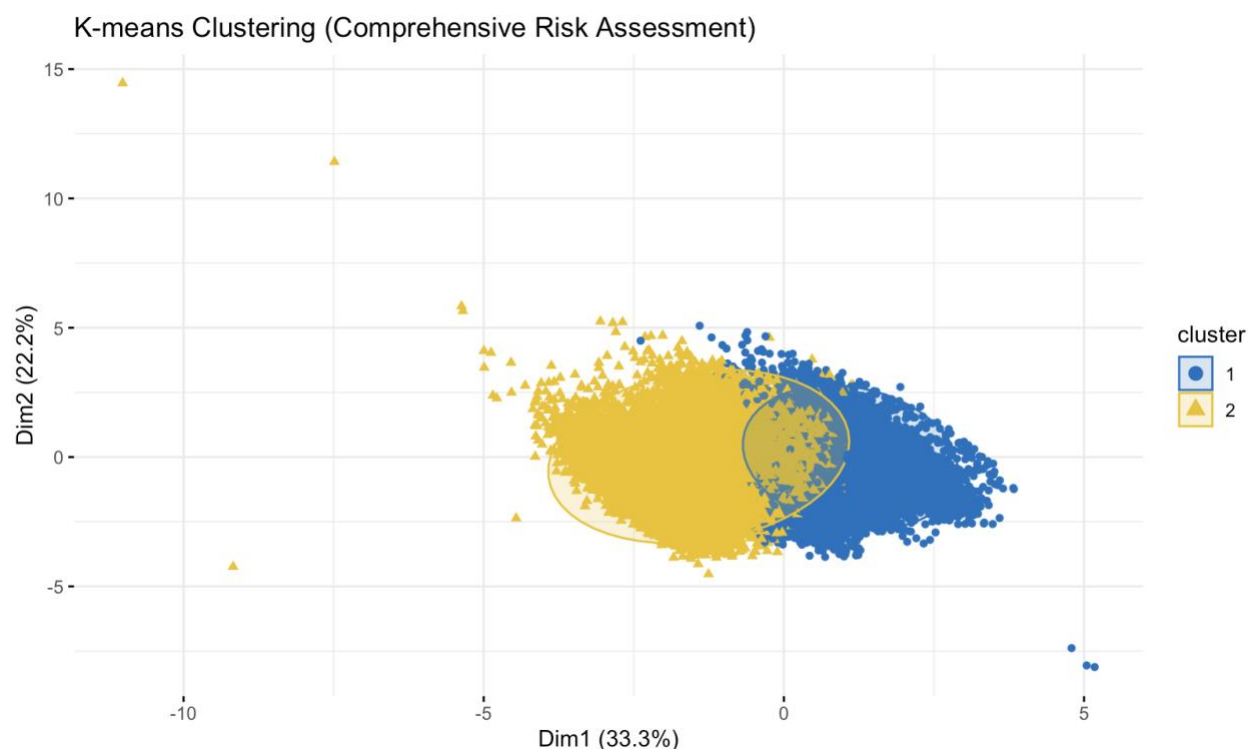


Figure 35: K-means Clustering (Comprehensive Risk, K=2)

Compressing the comprehensive feature set into two clusters reduces the overall silhouette by approximately 0.179. However, this approach results in uneven cluster quality, with the C2 silhouette being around 0.361 and the C1 silhouette being only 0.040. This suggests that a two-part split oversimplifies the portfolio's heterogeneity. The cluster centers indicate that the two groups are partially separated by loan age, macro exposure, and credit quality. One cluster consists of older loans with worse macro exposure and slightly higher LTV, while the other contains younger loans with stronger balance and FICO signals. Despite this, the weak silhouette for one cluster implies that this binary view conflates distinct behaviors that would be better preserved in a three-tier solution. In practical terms, while the K=2 comprehensive model can support a coarse, high-level division that is useful for some governance purposes, such as binary “watch/normal” routing, it is less effective for precise risk-based pricing, securitization tranching, or payoff forecasting tasks. Analytically, the low separation warns against relying on two-cluster comprehensive segmentation for fine-grained policy decisions. Instead, it suggests using K=3 (or the time-based K=2 for lifecycle tasks) as the more useful operational choice.

In summary, the clustering experiments provide a valuable toolkit that aligns directly with your five business goals and the analytical constraints you outlined. The Comprehensive Risk (K=3) and Full Variable (K=3) clusters offer the three-tier structure essential for risk-based pricing, underwriting segmentation, and securitization tranching. The Subset Variables (K=3) cluster isolates a high-confidence, high-risk cohort, making it ideal for immediate intervention and model-driven collections. The Time-Based (K=2) segmentation stands out for its unique power in payoff-timing models and cash-flow forecasting. The Borrower-Profile (K=2) clusters provide transparent, explainable buckets that facilitate regulatory and underwriting workflows.

Additionally, the results highlight areas where further analytical attention is required, such as small or low-silhouette clusters like Full Variable C3 or Comprehensive K=2 C1. These clusters necessitate specialized follow-up, including separate survival modeling, outlier analysis, or the incorporation of additional features. Operationally, it is advisable to utilize the high-silhouette clusters for automated policies, while treating low-silhouette groups as candidates for bespoke strategies and more in-depth modeling. Analytically, these clusters validate the necessity of integrating panel/time methods with macro variables, guide the focus of feature engineering (e.g., LTV_time, uer_time, loan_age), and assist in managing class imbalance by providing targeted oversampling or stratified validation within problematic segments.

11. Conclusion

This study presents a comprehensive and operationally robust framework for enhancing mortgage risk management by integrating classical statistical models, machine learning classifiers, and unsupervised clustering. Across all supervised models, predictive performance consistently demonstrated strong discriminatory power, with AUC values ranging from approximately 0.84 to 0.90 and stable performance across training, validation, and test sets. Gradient boosting and bagging models emerged as the most effective classification approaches, achieving the highest F1-scores and the best balance between sensitivity and specificity, particularly at the optimized 0.35 threshold. These models significantly improved the detection of high-risk borrowers compared to baseline logistic regression, directly supporting the business objective of enhancing default prediction accuracy by at least 20%. Fixed-effects logistic regression provided a reliable and interpretable baseline, while random-effects logistic regression introduced important structural realism by explicitly modeling time-based heterogeneity. Although random-effects models did not numerically surpass fixed-effects models, they produced more stable sensitivity across economic regimes, confirming the fundamental time-dependence of mortgage risk and validating the importance of panel-based modeling for long-horizon loan portfolios.

From an operational standpoint, the sensitivity-specificity trade-offs observed across various thresholds serve as a clear decision-making tool for management. At the 0.50 threshold, all models strongly prioritize specificity and precision, rendering them suitable for capital protection, securitization screening, and regulatory reporting where false positives are financially burdensome. Conversely, at the 0.35 threshold, sensitivity significantly increases (ranging from 0.78 to 0.82 for bagging and boosting), making these models more effective for early-warning systems, collections prioritization, and borrower outreach. This dual-threshold strategy enables the institution to dynamically adjust between conservative risk containment and aggressive early intervention modes in response to market conditions.

The unsupervised K-means clustering results provide critical strategic depth by revealing distinct structural segments within the portfolio that classification alone cannot capture. The Comprehensive Risk (K=3) and Full Variable (K=3) models successfully identified three economically significant risk tiers—low, moderate, and high risk—formed by the joint interactions between borrower credit quality, current loan leverage, macroeconomic exposure, and loan lifecycle dynamics. These clusters directly enable risk-based pricing, tiered underwriting policies, and tranche-based securitization design. The Time-Based (K=2) clustering emerged as the most effective segmentation structure and is particularly valuable for improving loan payoff timing forecasts, thereby enhancing cash-flow predictability and strengthening hedging strategies. Additionally, the Borrower Risk Profile clustering offers a transparent, regulator-friendly segmentation system that aligns closely with underwriting governance and explainable credit decisioning.

Crucially, the models also provide compelling evidence that macroeconomic stress—particularly unemployment, HPI decline, and elevated current LTV—systematically amplifies default risk across various modeling frameworks. This consistent signal across logistic regression, ensemble models, and clustering validates the portfolio's sensitivity to economic

cycles and establishes a robust foundation for scenario-based stress testing and capital planning. Consequently, institutions can now simulate macroeconomic shocks and directly map their effects onto the most vulnerable clusters and probability bands, thereby enhancing portfolio resilience in adverse conditions.

From a practical business perspective, the findings indicate a multi-layered deployment strategy. Gradient boosting and bagging should be the primary early-warning engines due to their superior sensitivity and F1 performance. Random-effects logistic regression should serve as the structural benchmark model for regulatory transparency and longitudinal risk monitoring. Comprehensive K-means clusters should guide risk-tier pricing, underwriting segmentation, and securitization structuring. Time-based clusters should anchor payoff forecasting and liquidity planning. High-risk cluster members should be routed into proactive intervention pipelines, including tighter credit overlays, accelerated borrower contact, and targeted refinancing or modification offers.

In conclusion, this integrated modeling framework significantly enhances all five business objectives: it improves default prediction accuracy, enhances payoff forecasting precision, enables statistically defensible risk-based pricing, deepens macroeconomic stress testing capabilities, and translates predictive outputs into actionable policy measures. Analytically, it addresses the challenges posed by panel data, class imbalance, collinearity, and time-dependent risk by selecting the appropriate modeling class for each structural complexity. Consequently, these models transform the portfolio from a reactive risk management system into a forward-looking, scenario-aware, and policy-driven decision platform capable of maintaining profitability and stability throughout economic cycles.